

Time-Optimal Candidate Heliocentric Transfers With a Constant-Power, Variable- I_{sp} Engine

Jan Olšina

Independent Researcher, Prague, Czech Republic

email: jan.olsina@gmail.com, ORCID: 0009-0003-9816-3958.

Abstract—We study time-optimal candidate heliocentric transfers for a spacecraft propelled by an electric thruster that supplies constant useful jet power P while continuously varying its exhaust speed (variable I_{sp}). The vehicle is assumed to depart from and arrive on circular heliocentric orbits, so that the initial and final velocities match the local circular velocities at the respective radii. First, we derive the analytic one-dimensional, gravity-free brachistochrone and characterize the normal-PMP structure produced by a finite exhaust-speed ceiling. The ceiling introduces maximum-exhaust-speed boundary arcs and a central coast; every finite-switch candidate is an endpoint truncation of an interior-boost–boundary-boost–coast–boundary-brake–interior-brake–boundary-brake sequence. Although the convexified first-order equations also admit a coast/boundary sliding extremal, an exchange argument proves that this relaxed arc cannot belong to a true time-minimizing gravity-free trajectory. Next, we formulate the full planar Sun-field optimal-control problem, derive two closed-form first integrals, and show that the indirect formulation reduces the orbital/costate shooting dynamics to seven differential equations, with propellant consumption enforced by an additional quadrature or endpoint constraint. Finally, we present a practical numerical continuation strategy that obtains a coarse feasible endpoint via global optimization and then refines it by homotopy and Powell’s local solver. Numerical examples for a 1 GW useful jet-power engine with an initial/dry mass of 3000 t $\rightarrow$ 1000 t produce Earth–Jupiter-class PMP-extremal transfer candidates with flight times ranging from roughly 170 days for short-arc geometries up to about 215 days for long-arc geometries; the longer-arc candidates highlighted in the figures commonly exploit a low-perihelion Oberth-like pass. All results are subject to the idealizations of the model.

I. INTRODUCTION

While electric propulsion (EP) is traditionally valued for its propellant efficiency, future ambitious endeavors such as interplanetary logistics networks or rapid sample return missions will demand not just efficiency but *speed* [1], [2], [3]. For such missions, the design focus shifts from optimizing propellant mass to minimizing flight time, making the spacecraft’s power system the primary limiting factor [4]. Reducing transfer times from years to months propels interest in high-power propulsion systems, with gigawatt-class architectures representing a plausible long-term goal. Several pathways to this power scale are being explored, including advanced Nuclear Electric Propulsion (NEP) [5], [6], next-generation deployable solar arrays [7], and beamed-energy concepts [8]. A common feature of these technologies is the ability to provide nearly constant useful propulsion power over long thrust arcs. This constant-power paradigm

fundamentally alters the classic rocketry framework. Instead of the fixed exhaust velocity (v_e) of the Tsiolkovsky equation [9], the exhaust speed becomes a key control variable, allowing the engine to actively trade thrust for specific impulse. The Variable Specific Impulse Magnetoplasma Rocket (VASIMR) is an exemplary technology demonstrating this “constant-power throttling” over a wide operating range [10]. Such throttleability is compelling for the fast transfers that motivate our study, as recent optimization work confirms that varying the specific impulse during flight can significantly improve trip time [11]. In this paper, we develop the analytical and numerical machinery for constructing time-optimal transfer candidates in this high-power regime. We first derive an analytical solution for a 1D, gravity-free “brachistochrone” to build intuition. We then formulate the full planar Sun-field problem, derive two closed-form first integrals to simplify the dynamics, and present a robust numerical strategy to solve the resulting boundary-value problem. Our idealized examples for a gigawatt-scale engine demonstrate Earth–Jupiter-class transfer candidates in a few months, frequently exploiting a low-perihelion Oberth-like maneuver.

II. VARIABLE- I_{sp} ROCKET MODEL

Let $m(t)$ be the instantaneous mass ($m \geq m_{\text{dry}}$) and $v_e(t)$ the exhaust speed. The propellant mass flow rate is the positive quantity $\dot{m}_f = -\dot{m}$. Neglecting plume inefficiencies, the useful jet power P (taken to be constant) and thrust magnitude T are given by [12]:

$$P = \frac{1}{2}(-\dot{m})v_e^2, \quad T = m a = (-\dot{m})v_e, \quad (1)$$

where a is the resulting thrust-induced acceleration magnitude.¹ Eliminating v_e gives a direct relationship between the mass flow rate and the vehicle’s acceleration:

$$\dot{m} = -\frac{m^2 a^2}{2P}. \quad (2)$$

This equation forms the basis of our optimal control problem.

¹In the vector formulation (Sec. V), $a = \|\mathbf{a}_T\|$ is the thrust-acceleration magnitude. In the free-space derivation below $a(t)$ is used as a signed scalar; since (2) depends on a^2 , the sign convention is immaterial for the propellant balance.

III. EXACT SOLUTION FOR THE FREE-SPACE CASE

$$(\mu = 0)$$

With solar gravity absent ($\mu = 0$), the translational degrees of freedom decouple. The problem simplifies to finding the minimum-time trajectory for a 1D drive from an initial state $(x, v) = (0, 0)$ at $t = 0$ to a final state $(x, v) = (L, 0)$ at the final time t_f .

A. The Variational Method and the Hamiltonian

To find the optimal control history—in this case, the acceleration profile $a(t)$ —we use the calculus of variations, specifically Pontryagin's Maximum Principle (PMP) [13]. We use the normal maximization convention, in which the Hamiltonian is written as

$$\mathcal{H} = \boldsymbol{\lambda}^\top \mathbf{f}(\mathbf{x}, \mathbf{u}) - L. \quad (3)$$

For a minimum-time problem, the performance index is

$$J = \int_0^{t_f} 1 \, dt,$$

so the running cost is $L = 1$.² For an autonomous problem with free final time, transversality implies $\mathcal{H} \equiv 0$ along the optimal trajectory.

Here:

- L is the **running cost**. For the minimum-time problem considered here, $L = 1$.
- $\mathbf{x}$ is the **state vector**, containing the variables that describe the system.
- $\mathbf{u}$ is the **control vector**, containing the variables we can change to steer the system.
- $\mathbf{f}(\mathbf{x}, \mathbf{u}) = \dot{\mathbf{x}}$ is the vector of **state dynamics**.
- $\boldsymbol{\lambda}$ is the **costate vector**, a set of auxiliary variables introduced by the PMP.

B. Hamiltonian Construction for the Free-Space Case

We apply the principle from Section III-A. First, we identify the components of the 1D problem:

- **State vector:** $\mathbf{x} = (x, v, m)^\top$.
- **Control variable:** $u = a(t)$.
- **State dynamics** $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, u)$:

$$\dot{x} = v, \quad (4)$$

$$\dot{v} = a, \quad (5)$$

$$\dot{m} = -\frac{m^2 a^2}{2P}. \quad (6)$$

- **Costate vector:** $\boldsymbol{\lambda} = (\lambda_x, \lambda_v, \lambda_m)^\top$.

The term $\boldsymbol{\lambda}^\top \dot{\mathbf{x}}$ in the Hamiltonian expands to:

$$\boldsymbol{\lambda}^\top \dot{\mathbf{x}} = \lambda_x \dot{x} + \lambda_v \dot{v} + \lambda_m \dot{m} = \lambda_x v + \lambda_v a - \lambda_m \frac{m^2 a^2}{2P}.$$

Subtracting the running cost $L = 1$, according to the convention in (3), gives the full Hamiltonian

$$\mathcal{H} = \lambda_x v + \lambda_v a - \lambda_m \frac{m^2 a^2}{2P} - 1. \quad (7)$$

²Many modern aerospace optimal-control texts write $\mathcal{H} = \boldsymbol{\lambda}^\top \mathbf{f} + L$ and minimize; that convention differs from (3) by the sign of all costates. The present maximization form follows Pontryagin's original formulation [13].

From this Hamiltonian, we derive the costate (adjoint) equations, given by $\dot{\boldsymbol{\lambda}} = -\partial \mathcal{H} / \partial \mathbf{x}$:

$$\dot{\lambda}_x = -\frac{\partial \mathcal{H}}{\partial x} = 0, \quad (8a)$$

$$\dot{\lambda}_v = -\frac{\partial \mathcal{H}}{\partial v} = -\lambda_x, \quad (8b)$$

$$\dot{\lambda}_m = -\frac{\partial \mathcal{H}}{\partial m} = \lambda_m \frac{ma^2}{P}. \quad (8c)$$

C. Optimal Control and First Integrals

According to the PMP, the optimal control $a(t)$ must maximize the Hamiltonian at all times. We find it by setting $\partial \mathcal{H} / \partial a = 0$:

$$\frac{\partial \mathcal{H}}{\partial a} = \lambda_v - \lambda_m \frac{m^2 a}{P} = 0 \implies a(t) = \frac{P \lambda_v(t)}{\lambda_m(t) m^2(t)}. \quad (9)$$

The stationary point is a maximum when $\lambda_m > 0$ (equivalently, $C_m \equiv \lambda_m m^2 > 0$), making the Hamiltonian concave in a . From the costate dynamics, we can extract two integrals of motion. First, (8a) directly implies:

$$\lambda_x(t) = C_1 = \text{const.} \quad (10)$$

Substituting this into (8b) and integrating gives:

$$\lambda_v(t) = -C_1 t + C_2, \quad (11)$$

showing that the velocity costate is a linear function of time.

A second, less obvious, integral is found by examining the time derivative of the product $\lambda_m m^2$:

$$\begin{aligned} \frac{d}{dt}(\lambda_m m^2) &= \dot{\lambda}_m m^2 + \lambda_m (2m \dot{m}) \\ &= \left(\lambda_m \frac{ma^2}{P} \right) m^2 + \lambda_m \left(2m \left(-\frac{m^2 a^2}{2P} \right) \right) \\ &= \frac{\lambda_m m^3 a^2}{P} - \frac{\lambda_m m^3 a^2}{P} = 0. \end{aligned} \quad (12)$$

Thus, the product $\lambda_m(t) m^2(t)$ is constant throughout the trajectory:

$$\lambda_m(t) m^2(t) = C_m = \text{const.} \quad (13)$$

Substituting these integrals into the control law (9), we discover that the optimal acceleration profile must be linear in time:

$$a(t) = \frac{P}{C_m} (-C_1 t + C_2) = K_1 t + K_0, \quad (14)$$

where $K_1 = -PC_1/C_m$ and $K_0 = PC_2/C_m$ are constants.

D. Deriving the Trajectory

We now determine the constants K_1 and K_0 by applying the boundary conditions. Let the final time be t_f . First, we integrate $a(t)$ to find the velocity $v(t)$:

$$v(t) = \int_0^t a(\tau) d\tau + v(0) = \frac{1}{2} K_1 t^2 + K_0 t. \quad (15)$$

Applying the final velocity condition $v(t_f) = 0$:

$$\frac{1}{2} K_1 t_f^2 + K_0 t_f = 0 \implies K_0 = -\frac{1}{2} K_1 t_f. \quad (16)$$

Substituting this back into $a(t)$ reveals its symmetric nature, a characteristic feature of such time-optimal "there-and-back" problems:

$$a(t) = K_1 t - \frac{1}{2} K_1 t_f = K_1 (t - t_f/2). \quad (17)$$

Next, we integrate $v(t)$ to find the position $x(t)$:

$$x(t) = \int_0^t v(\tau) d\tau + x(0) = \frac{1}{6} K_1 t^3 + \frac{1}{2} K_0 t^2. \quad (18)$$

Applying the final position condition $x(t_f) = L$:

$$L = \frac{1}{6} K_1 t_f^3 + \frac{1}{2} K_0 t_f^2. \quad (19)$$

Substituting $K_0 = -\frac{1}{2} K_1 t_f$ into the equation for L allows us to solve for K_1 :

$$\begin{aligned} L &= \frac{1}{6} K_1 t_f^3 + \frac{1}{2} (-\frac{1}{2} K_1 t_f) t_f^2 \\ &= (\frac{1}{6} - \frac{1}{4}) K_1 t_f^3 = -\frac{1}{12} K_1 t_f^3. \end{aligned} \quad (20)$$

This gives $K_1 = -12L/t_f^3$. Consequently, $K_0 = -\frac{1}{2} (-\frac{12L}{t_f^3}) t_f = 6L/t_f^2$. The optimal acceleration is therefore:

$$a(t) = \frac{12L}{t_f^3} \left(\frac{t_f}{2} - t \right). \quad (21)$$

This corresponds to an initial acceleration phase, followed by a symmetric braking phase, with $a(t)$ crossing zero at mid-time, $t = t_f/2$. By substituting the constants back and simplifying, we obtain the full trajectory:

$$v(t) = \frac{6Lt}{t_f^2} \left(1 - \frac{t}{t_f} \right), \quad (22)$$

$$x(t) = L \left[3 \left(\frac{t}{t_f} \right)^2 - 2 \left(\frac{t}{t_f} \right)^3 \right], \quad (23)$$

which describes a cubic-in-time "S-curve" path.

E. Mass Evolution and Minimum Time

The mass evolution is governed by the separable differential equation (6). Integrating gives:

$$\begin{aligned} \int_{m_i}^{m(t)} \frac{dm}{m^2} &= - \int_0^t \frac{a(\tau)^2}{2P} d\tau \\ \Rightarrow \frac{1}{m_i} - \frac{1}{m(t)} &= - \frac{1}{2P} \int_0^t a(\tau)^2 d\tau. \end{aligned} \quad (24)$$

The integral of $a(t)^2$ is:

$$\begin{aligned} \int_0^t a(\tau)^2 d\tau &= \int_0^t \left(\frac{12L}{t_f^3} \right)^2 \left(\frac{t_f}{2} - \tau \right)^2 d\tau \\ &= \frac{144L^2}{t_f^6} \left[-\frac{1}{3} \left(\frac{t_f}{2} - \tau \right)^3 \right]_0^t \\ &= \frac{48L^2}{t_f^6} \left(\frac{t_f^3}{8} - \left(\frac{t_f}{2} - t \right)^3 \right). \end{aligned} \quad (25)$$

Defining

$$f(s) = 4s^3 - 6s^2 + 3s, \quad (26)$$

the mass relation becomes:

$$\frac{1}{m(t)} - \frac{1}{m_i} = \frac{6L^2 f(t/t_f)}{P t_f^3}. \quad (27)$$

The total propellant consumed is found by setting $t = t_f$ (or $s = 1$). If the entire propellant reserve is used, we have $m(t_f) = m_{\text{dry}}$, which defines the minimum possible flight-time for the given distance L and mass fraction:

$$\Delta \equiv \frac{1}{m_{\text{dry}}} - \frac{1}{m_i} \quad (28)$$

calculated as

$$\Delta = \frac{6L^2 f(1)}{P T_{\min}^3} = \frac{6L^2}{P T_{\min}^3}. \quad (29)$$

Solving for $T_{\min}$ yields the final result:

$$T_{\min} = \left[\frac{6L^2}{P\Delta} \right]^{1/3}. \quad (30)$$

Equation (30) treats the dry mass as prescribed. If the propulsion hardware mass is instead allowed to scale with power, the same free-space benchmark can be extended by a simple dry-mass closure; the resulting scaling law is derived in Appendix A.

IV. FREE-SPACE TRANSFER WITH A FINITE EXHAUST-SPEED CEILING

The uncapped free-space solution of Sec. III requires

$$v_e(t) = \frac{2P}{m(t)|a(t)|}, \quad (31)$$

and therefore demands arbitrarily large exhaust speed when the affine acceleration passes through zero. We now impose a finite ceiling

$$v_e \leq V, \quad V \equiv v_{e,\text{max}}, \quad (32)$$

while retaining the two-mode engine model used throughout this section: when the engine is on, it supplies the fixed useful jet power P ; when it is off, the spacecraft coasts with $a = 0$ and $\dot{m} = 0$. We seek the minimum final time t_f subject to

$$\begin{aligned} x(0) &= 0, & v(0) &= 0, & m(0) &= m_i, \\ x(t_f) &= L, & v(t_f) &= 0, & m(t_f) &= m_{\text{dry}}. \end{aligned} \quad (33)$$

The resulting control set is nonconvex, so the capped solution is not obtained by merely truncating or uniformly scaling the uncapped affine-acceleration profile.

It is convenient to define the thrust and propellant-flow rate associated with full power at the exhaust-speed ceiling,

$$F \equiv \frac{2P}{V}, \quad \sigma \equiv \frac{2P}{V^2} = \frac{F}{V}. \quad (34)$$

During a powered interval,

$$v_e = \frac{2P}{m|a|} \leq V \iff |a| \geq \frac{F}{m}. \quad (35)$$

Consequently, the admissible signed accelerations are

$$\mathcal{U}(m) = \{0\} \cup \left(-\infty, -\frac{F}{m} \right] \cup \left[\frac{F}{m}, \infty \right). \quad (36)$$

The nonzero powered acceleration therefore has a *minimum* magnitude. Accelerations between zero and F/m cannot be produced at fixed power without violating the exhaust-speed ceiling; the engine must instead be switched off.

A. Pointwise Hamiltonian Maximization

The state equations are

$$\dot{x} = v, \quad (37a)$$

$$\dot{v} = a, \quad (37b)$$

$$\dot{m} = -\frac{m^2 a^2}{2P}, \quad (37c)$$

with the understanding that $a = 0$ denotes the coast mode. In the normal maximization convention, the Hamiltonian is

$$\mathcal{H} = \lambda_x v + \lambda_v a - \lambda_m \frac{m^2 a^2}{2P} - 1. \quad (38)$$

As in the uncapped problem,

$$\dot{\lambda}_x = 0, \quad \dot{\lambda}_v = -\lambda_x, \quad \dot{\lambda}_m = \lambda_m \frac{ma^2}{P} \quad (39)$$

on powered arcs, while $\dot{\lambda}_m = 0$ on a coast. Hence

$$C \equiv \lambda_m m^2 \quad (40)$$

is constant on every arc and across the switches. A finite pointwise maximum of (38) requires $C > 0$. Define the scalar primer acceleration

$$p(t) \equiv \frac{P\lambda_v(t)}{C} = p_0 - \omega t, \quad \omega \equiv \frac{P\lambda_x}{C}. \quad (41)$$

For a transfer from rest at $x = 0$ to rest at $x = L > 0$, the maximizing acceleration must be positive before it is negative. Since the sign of the maximizing powered acceleration is the sign of p , the primer must decrease through zero, and therefore $\omega > 0$.

The part of the Hamiltonian that depends on a is, apart from the positive factor C/P ,

$$\Phi(a) = pa - \frac{a^2}{2}. \quad (42)$$

Let

$$q(m) \equiv \frac{F}{m}. \quad (43)$$

If $|p| \geq q$, the unconstrained stationary point $a = p$ is feasible and maximizes Φ . If $|p| < q$, the best powered control is the boundary value $a = \text{sgn}(p)q$, whose Hamiltonian gain relative to coasting is

$$\Phi(\text{sgn}(p)q) = |p|q - \frac{q^2}{2}. \quad (44)$$

This gain is positive only when $|p| > q/2$. The exact pointwise maximizing law is therefore

$$a^*(t) = \begin{cases} p(t), & |p(t)| \geq q, \\ \text{sgn}(p(t))q, & q/2 < |p(t)| < q, \\ 0, & |p(t)| \leq q/2. \end{cases} \quad (45)$$

At $|p| = F/m$, the interior and boundary controls coincide. At $|p| = F/(2m)$, the maximum-exhaust-speed control and coasting have equal Hamiltonian. These equalities define the switching surfaces.

Three arc types result:

- On an *interior powered arc*, denoted $U_{\pm}$, the ceiling is inactive and

$$a = p, \quad v_e = \frac{2P}{m|p|} < V. \quad (46)$$

- On a *maximum-exhaust-speed boundary arc*, denoted $B_{\pm}$,

$$a = \pm \frac{F}{m}, \quad v_e = V, \quad \dot{m} = -\sigma. \quad (47)$$

Thus thrust and mass-flow rate are constant, while the acceleration magnitude increases as the vehicle mass decreases.

- On a *coast arc*, denoted C , $a = 0$ and m is constant.

The subscripts $+$ and $-$ indicate acceleration and braking, respectively.

B. Closed-Form Arc Relations

Because $\dot{p} = -\omega$, each arc can be integrated in closed form.

On an interior powered arc,

$$\frac{dm}{dp} = \frac{m^2 p^2}{2P\omega}, \quad (48)$$

so an arc beginning at (p_a, m_a) satisfies

$$\frac{1}{m(p)} = \frac{1}{m_a} + \frac{p_a^3 - p^3}{6P\omega}. \quad (49)$$

On either boundary arc,

$$\frac{dm}{dp} = \frac{\sigma}{\omega}, \quad m(p) = m_a + \frac{\sigma}{\omega}(p - p_a). \quad (50)$$

On a coast, m is constant while p continues to decrease linearly.

The corresponding state maps are also elementary. For an interior powered arc of duration τ beginning with (x_a, v_a, p_a) ,

$$p(\tau) = p_a - \omega\tau, \quad (51a)$$

$$v(\tau) = v_a + p_a\tau - \frac{1}{2}\omega\tau^2, \quad (51b)$$

$$x(\tau) = x_a + v_a\tau + \frac{1}{2}p_a\tau^2 - \frac{1}{6}\omega\tau^3. \quad (51c)$$

For a boundary arc with sign $\varepsilon = +1$ or -1 ,

$$m(\tau) = m_a - \sigma\tau, \quad (52a)$$

$$v(\tau) = v_a + \varepsilon V \ln\left(\frac{m_a}{m(\tau)}\right), \quad (52b)$$

$$x(\tau) = x_a + v_a\tau + \varepsilon V \left[\tau - \frac{m(\tau)}{\sigma} \ln\left(\frac{m_a}{m(\tau)}\right) \right]. \quad (52c)$$

A coast of duration τ has $v(\tau) = v_a$ and $x(\tau) = x_a + v_a\tau$.

C. Ordering of the Arcs

The switching law (45) determines the arc ordering; it is not an independent ansatz. Introduce the normalized primer magnitude

$$z(t) \equiv \frac{|p(t)|m(t)}{F}. \quad (53)$$

The interior, boundary, and coast regions are $z \geq 1$, $1/2 < z < 1$, and $z \leq 1/2$, respectively.

While $p > 0$, $z = pm/F$. Along an interior arc, a boundary arc, and a coast, respectively,

$$\dot{z}_U = -\frac{1}{F} \left(\omega m + \frac{m^2 p^3}{2P} \right) < 0, \quad (54a)$$

$$\dot{z}_B = -\frac{1}{F} (\omega m + \sigma p) < 0, \quad (54b)$$

$$\dot{z}_C = -\frac{\omega m}{F} < 0. \quad (54c)$$

Thus the accelerating side can only have the order

$$[U_+] B_+ C, \quad (55)$$

where brackets denote an optional arc. An initial coast would leave the spacecraft at rest while increasing the transfer time, and is therefore excluded. The boundary arc B_+ cannot be skipped: a continuous trajectory from $z > 1/2$ to the coast region must pass through $1/2 < z < 1$. Moreover, for every finite V , the central coast has positive duration because p crosses zero while $F/(2m) > 0$.

On the braking side, write $u \equiv -p > 0$, so that $\dot{u} = \omega$ and $z = um/F$. The coast necessarily ends at $z = 1/2$, after which a regular trajectory enters B_- . On that boundary arc,

$$\dot{z}_B = \frac{1}{F} (\omega m - \sigma u). \quad (56)$$

The quantity z can initially increase, reach a single maximum, and then decrease. If it reaches $z = 1$, the trajectory enters U_- . On U_- , Eq. (49) gives

$$\frac{1}{m(u)} = \frac{1}{m_4} + \frac{u^3 - u_4^3}{6P\omega}, \quad (57)$$

where (u_4, m_4) is the $B_- \rightarrow U_-$ switch and $u_4 = F/m_4$. Differentiating $z(u) = um(u)/F$ shows that the sign of dz/du is the sign of

$$\frac{1}{m_4} - \frac{u_4^3}{6P\omega} - \frac{u^3}{3P\omega}, \quad (58)$$

which decreases strictly with u . Hence z has at most one maximum on U_- , and the trajectory can leave U_- at most once. If it does, it re-enters B_- at $z = 1$ and subsequently moves toward the $z = 1/2$ switching surface.

It follows that every regular normal extremal with finitely many switches is an endpoint truncation of

$$[U_+] B_+ C B_- [U_- [B_-]]. \quad (59)$$

Equivalently, the six finite regular topologies are

$$\begin{aligned} B_+ C B_-, & \quad U_+ B_+ C B_-, \\ B_+ C B_- U_-, & \quad U_+ B_+ C B_- U_-, \\ B_+ C B_- U_- B_-, & \quad U_+ B_+ C B_- U_- B_-. \end{aligned} \quad (60)$$

The final time may occur in the first braking boundary arc, in the braking interior arc, or in the second braking boundary arc. A zero-length endpoint arc represents only the limiting case in which the endpoint lies on a switching surface; an endpoint may also lie strictly inside a boundary region.

A finite sequence containing repeated, separated coast intervals, such as $B_- C B_- C B_-$, is not a regular finite-switch extremal. When a descending braking branch reaches $z = 1/2$, coasting makes z increase, whereas full-power boundary

operation makes z decrease. The two classical vector fields therefore point toward the switching surface from opposite sides, and the pointwise selector has no continuation consisting of positive-duration alternating coast and boundary arcs.

D. Relaxed Sliding Extremal and Its Exclusion

Convexifying the on-off control set produces a formal sliding extremal on the coast-boundary switching surface

$$z = \frac{1}{2}, \quad u = \frac{F}{2m}. \quad (61)$$

Let $\delta \in [0, 1]$ denote the local fraction of time for which the engine operates at full power with $v_e = V$. The averaged coast/boundary dynamics are

$$\bar{a} = -\delta \frac{F}{m}, \quad \dot{m} = -\delta \sigma. \quad (62)$$

Requiring $\dot{z} = 0$ on Eq. (61) gives

$$\delta(m) = \frac{2\omega m^2}{\sigma F}, \quad 0 < \delta \leq 1, \quad (63)$$

and therefore

$$\bar{a} = -\frac{2\omega m}{\sigma}, \quad \dot{m} = -\frac{2\omega m^2}{F}. \quad (64)$$

Under the strict full-power/on-off model, this control is the weak limit of increasingly rapid alternation between C and B_- . If continuous power throttling is admitted, the same averaged equations are obtained by operating at $v_e = V$ with useful power δP .

The sliding construction satisfies the convexified pointwise PMP condition, but it is not time minimizing in the gravity-free problem. This follows from an exchange argument that applies to every mixture of coasting and maximum-exhaust-speed braking, including the chattering limit.

Consider an interval beginning with state (x_a, v_a, m_a) on which the only controls are coast and B_- . Write $\eta(t) \in [0, 1]$ for the full-power duty fraction, where $\eta = 0$ is coast and $\eta = 1$ is full-power boundary braking. Then

$$\dot{m} = -\eta \sigma, \quad \dot{v} = -\eta \frac{F}{m}. \quad (65)$$

Introduce the accumulated full-power time

$$s(t) = \int_{t_a}^t \eta(\tau) d\tau. \quad (66)$$

Equations (65) integrate to

$$m(t) = m_a - \sigma s(t), \quad v(t) = v_a - V \ln \left(\frac{m_a}{m(t)} \right). \quad (67)$$

Thus the terminal mass and velocity depend only on the total full-power time $s(t_b)$, not on the placement of the coast intervals.

The effect on distance is ordered. Consider two adjacent intervals consisting of a boundary burn followed by a coast. Interchanging their order leaves the burn trajectory, elapsed time, terminal mass, and terminal velocity unchanged, because the preceding coast changes neither mass nor velocity. If the

burn reduces the velocity by $\Delta v > 0$ and the coast duration is $\tau_c > 0$, the coast-then-burn ordering travels farther by

$$\Delta x = \Delta v \tau_c > 0. \quad (68)$$

Repeatedly applying this interchange moves every coast interval ahead of every boundary-burn interval. Consequently, among all coast/ B_- schedules having the same total duration and the same total full-power time, the ordering

$$C \ B_- \quad (69)$$

travels the greatest distance. The result extends to a fractional control $0 < \eta < 1$ by approximation with on-off pulses, and hence applies to the sliding extremal.

The distance improvement can be converted into a strict time improvement. Let b and c be the total full-power and coast durations of a nontrivial mixed interval, and let $D_B(b)$ be the distance traveled by the uninterrupted boundary burn of duration b beginning from (v_a, m_a) . Inserting coast intervals into this burn does not change the powered state history; it only adds distance at the velocity at which each coast is inserted. Therefore the original mixed interval has distance

$$D_{\text{mix}} = D_B(b) + \int_{\text{coast}} v \, dt, \quad (70)$$

whereas the sorted coast-then-burn interval has distance

$$D_{CB} = D_B(b) + v_a c. \quad (71)$$

On a feasible terminal braking interval, $0 \leq v(t) \leq v_a$. Since velocity decreases during braking,

$$D_B(b) \leq D_{\text{mix}} < D_{CB} \quad (72)$$

whenever a positive amount of coast occurs after braking has begun. Define

$$c' \equiv \frac{D_{\text{mix}} - D_B(b)}{v_a}. \quad (73)$$

Then $0 \leq c' < c$, and a coast of duration c' followed by the same uninterrupted boundary burn reaches exactly the same final position, velocity, and mass in the strictly shorter time $b + c'$. Coast at zero velocity is even more directly removable. Hence no mixed coast/ B_- interval, no chattering limit, and no continuously throttled realization of Eq. (64) can be part of a true time-minimizing trajectory.

The first-order sliding extremal must therefore be treated as a nonoptimal artifact of convexification, not as an additional admissible optimal topology. A genuine gravity-free optimum may terminate on a braking boundary arc, but it must do so before a descending B_- branch reaches $z = 1/2$. All braking after the central coast is contained in one connected powered sequence,

$$B_- [U_- [B_-]], \quad (74)$$

with endpoint truncation allowed on any displayed arc. Together with Eq. (55), this leaves precisely the six regular topologies listed in Eq. (60). These are necessary normal-PMP candidates; the PMP alone does not establish sufficiency among them.

E. Switching Conditions on the Braking Side

Let M be the constant mass on the central coast and define

$$p_2 \equiv \frac{F}{2M}, \quad \chi \equiv \frac{\sigma F}{\omega M^2}. \quad (75)$$

The coast ends at $(u, m) = (p_2, M)$. If a full-power braking boundary arc is followed, then

$$u(m) = p_2 + \frac{\omega}{\sigma}(M - m), \quad (76)$$

and, with $y = m/M$,

$$z_B(y) = \frac{y}{2} + \frac{y(1-y)}{\chi}. \quad (77)$$

At the coast exit,

$$\dot{z}_B = \frac{\omega M}{F} \left(1 - \frac{\chi}{2}\right). \quad (78)$$

Thus a positive-length regular $C \rightarrow B_-$ switch exists only for $\chi < 2$. For $\chi \geq 2$, the boundary control immediately enters the coast-maximizing side of the switching surface. Such a trial extremal has no regular braking continuation and, by the exchange argument of the preceding subsection, cannot be completed optimally by a relaxed coast/boundary mixture; it must be rejected.

A positive-length U_- arc occurs only if z_B rises above unity. Solving $z_B = 1$ gives

$$\frac{m_4}{M} = \frac{2 + \chi + \sqrt{\chi^2 - 12\chi + 4}}{4}, \quad (79)$$

where the larger root is the first crossing encountered as the mass decreases. The boundary arc reaches $z = 1$ only when

$$\chi \leq \chi_c, \quad \chi_c \equiv 6 - 4\sqrt{2}. \quad (80)$$

At $\chi = \chi_c$, the boundary arc is tangent to $z = 1$ and the U_- arc has zero duration. For $\chi_c < \chi < 2$, no interior braking arc occurs; if the transfer continues far enough, the first B_- arc returns to $z = 1/2$ at

$$m_s = \frac{\chi}{2}M, \quad (81)$$

where the descending boundary branch again reaches the coast-boundary switching surface. A true minimum-time candidate must reach its terminal state no later than this point; a trial requiring $m_{\text{dry}} < m_s$ is rejected.

When $\chi < \chi_c$ and the trajectory enters U_- at $(u_4, m_4) = (F/m_4, m_4)$, the interior arc eventually returns to $z = 1$ if it is continued sufficiently far. Define

$$\beta \equiv \frac{F^3}{6P\omega m_4^2} = \frac{\sigma F}{3\omega m_4^2}. \quad (82)$$

The nontrivial return mass $m_5 < m_4$ follows from Eqs. (57) and $u = F/m$:

$$\frac{m_5}{m_4} = \frac{\beta + \sqrt{4\beta - 3\beta^2}}{2(1 - \beta)}. \quad (83)$$

The first $B_- \rightarrow U_-$ crossing lies on the increasing branch of z , which implies $0 < \beta < 1/3$ and makes (83) well defined. If $m_{\text{dry}} \geq m_4$, the transfer terminates on the first B_- arc before or at entry to U_- . If $m_5 \leq m_{\text{dry}} < m_4$, it terminates on U_-

before or at the return switch. Only when $m_{\text{dry}} < m_5$ does the trajectory re-enter B_- .

For the second boundary arc, set

$$\kappa \equiv \frac{\sigma F}{\omega m_5^2} > 1. \quad (84)$$

The mass at which this arc reaches $z = 1/2$ is

$$\frac{m_6}{m_5} = \frac{\kappa + 1 - \sqrt{\kappa^2 + 1}}{2}. \quad (85)$$

A transfer with $m_{\text{dry}} \geq m_6$ can terminate on this second B_- arc. A trial requiring $m_{\text{dry}} < m_6$ would reach the coast-boundary switching surface before the dry-mass endpoint and is therefore rejected rather than continued by a relaxed arc.

F. Exact Reduction to Two Endpoint Equations

The affine primer permits a compact formulation that covers every regular topology. Let the terminal primer be $p(t_f) = -u_f$, with $u_f > 0$. Since $dp/dt = -\omega$,

$$t_f = \frac{p_0 + u_f}{\omega}. \quad (86)$$

The zero-final-velocity condition is

$$0 = \int_0^{t_f} a \, dt = \frac{1}{\omega} \int_{-u_f}^{p_0} a^*(p, m(p)) \, dp. \quad (87)$$

Furthermore,

$$\begin{aligned} L &= \int_0^{t_f} v(t) \, dt = \int_0^{t_f} (t_f - t) a(t) \, dt \\ &= \frac{1}{\omega^2} \int_{-u_f}^{p_0} (p + u_f) a^*(p, m(p)) \, dp. \end{aligned} \quad (88)$$

The free-final-time condition $\mathcal{H} = 0$ fixes the remaining positive costate scale C/P after the primer trajectory has been determined; it does not introduce an additional state-shape equation.

The term proportional to u_f vanishes by (87), yielding

$$L = \frac{1}{\omega^2} \int_{-u_f}^{p_0} p a^*(p, m(p)) \, dp. \quad (89)$$

For prescribed $(P, V, L, m_i, m_{\text{dry}})$, one may choose the coast mass M and primer slope ω as the two unknowns. The positive-side switching law determines p_0 ; forward propagation of the regular braking-side switching law to $m = m_{\text{dry}}$ determines u_f . Equations (87) and (89) then provide two scalar residual equations for (M, ω) . All arc integrations are elementary by Eqs. (49)–(52c). A trial pair (M, ω) is rejected if the braking trajectory reaches the descending coast-boundary surface $z = 1/2$ before it reaches $m = m_{\text{dry}}$, because continuation by the relaxed sliding extremal is not time minimizing.

The accelerating branch is obtained directly from M and ω . If the boundary arc extends to the initial mass, then

$$p_0 = p_2 + \frac{\omega}{\sigma}(m_i - M), \quad \frac{p_0 m_i}{F} \leq 1, \quad (90)$$

and the initial topology is B_+ . Otherwise an interior arc precedes B_+ . Let m_1 denote the $U_+ \rightarrow B_+$ switching mass. Then

$$m_1 - M = \frac{\sigma F}{\omega} \left(\frac{1}{m_1} - \frac{1}{2M} \right), \quad (91)$$

so, with $r_1 = m_1/M$,

$$r_1 = \frac{2 - \chi + \sqrt{\chi^2 + 12\chi + 4}}{4}. \quad (92)$$

The initial primer is then

$$p_0 = \left[\left(\frac{F}{m_1} \right)^3 + 6P\omega \left(\frac{1}{m_1} - \frac{1}{m_i} \right) \right]^{1/3}. \quad (93)$$

The condition $m_1 < m_i$ is equivalent to a positive-length U_+ arc.

G. Regular Five-Arc Candidate

The most direct analogue of the uncapped solution is the regular five-arc topology

$$U_+ \, B_+ \, C \, B_- \, U_-. \quad (94)$$

It applies when

$$\begin{aligned} m_i > m_1 > M > m_4 > m_{\text{dry}}, \quad \chi < \chi_c, \\ \frac{u_f m_{\text{dry}}}{F} \geq 1. \end{aligned} \quad (95)$$

where the last inequality states that the dry-mass endpoint is reached before the braking interior arc returns to B_- . Define

$$p_1 \equiv \frac{F}{m_1}, \quad p_2 \equiv \frac{F}{2M}, \quad u_4 \equiv \frac{F}{m_4}. \quad (96)$$

The boundary matching conditions are

$$m_1 - M = \frac{\sigma F}{\omega} \left(\frac{1}{m_1} - \frac{1}{2M} \right), \quad (97a)$$

$$M - m_4 = \frac{\sigma F}{\omega} \left(\frac{1}{m_4} - \frac{1}{2M} \right). \quad (97b)$$

Their physical roots are

$$\frac{m_1}{M} = \frac{2 - \chi + \sqrt{\chi^2 + 12\chi + 4}}{4}, \quad (98a)$$

$$\frac{m_4}{M} = \frac{2 + \chi + \sqrt{\chi^2 - 12\chi + 4}}{4}. \quad (98b)$$

The interior-arc mass integrals give

$$p_0 = \left[p_1^3 + 6P\omega \left(\frac{1}{m_1} - \frac{1}{m_i} \right) \right]^{1/3}, \quad (99a)$$

$$u_f = \left[u_4^3 + 6P\omega \left(\frac{1}{m_{\text{dry}}} - \frac{1}{m_4} \right) \right]^{1/3}. \quad (99b)$$

The zero-final-velocity condition is the sum of the velocity changes on the four powered arcs,

$$\frac{p_0^2 - p_1^2}{2\omega} + V \ln\left(\frac{m_1}{M}\right) - V \ln\left(\frac{M}{m_4}\right) - \frac{u_f^2 - u_4^2}{2\omega} = 0. \quad (100)$$

For the distance equation, define

$$\mathcal{I}_{U+} = \frac{p_0^3 - p_1^3}{3}, \quad (101a)$$

$$\mathcal{I}_{U-} = \frac{u_f^3 - u_4^3}{3}, \quad (101b)$$

$$\mathcal{I}_{B+} = \frac{F\omega}{\sigma} \left[p_2 \ln\left(\frac{m_1}{M}\right) + \frac{\omega}{\sigma} \left(m_1 - M - M \ln\left(\frac{m_1}{M}\right) \right) \right], \quad (101c)$$

$$\mathcal{I}_{B-} = \frac{F\omega}{\sigma} \left[p_2 \ln\left(\frac{M}{m_4}\right) + \frac{\omega}{\sigma} \left(M \ln\left(\frac{M}{m_4}\right) - (M - m_4) \right) \right]. \quad (101d)$$

Equation (89) becomes

$$L = \frac{1}{\omega^2} (\mathcal{I}_{U+} + \mathcal{I}_{B+} + \mathcal{I}_{B-} + \mathcal{I}_{U-}). \quad (102)$$

Equations (100) and (102) are two nonlinear algebraic equations for (M, ω) . Once they are solved, all masses, primer values, switching times, and states follow algebraically. In particular,

$$t_1 = \frac{p_0 - p_1}{\omega}, \quad (103a)$$

$$t_2 = \frac{p_0 - p_2}{\omega}, \quad (103b)$$

$$t_3 = \frac{p_0 + p_2}{\omega}, \quad (103c)$$

$$t_4 = \frac{p_0 + u_4}{\omega}, \quad (103d)$$

$$t_f = \frac{p_0 + u_f}{\omega}. \quad (103e)$$

The acceleration profile is

$$a(t) = \begin{cases} p_0 - \omega t, & 0 \leq t < t_1, \\ F/m(t), & t_1 \leq t < t_2, \\ 0, & t_2 \leq t < t_3, \\ -F/m(t), & t_3 \leq t < t_4, \\ p_0 - \omega t, & t_4 \leq t \leq t_f. \end{cases} \quad (104)$$

Unlike the scaled boost-coast-brake approximation, this profile is generally not symmetric in time. The central coast is symmetric in primer space, extending from $p = p_2$ to $p = -p_2$, but the mass loss makes the accelerating and braking powered arcs unequal.

H. Limiting Behavior and Scope

As $V \rightarrow \infty$, both F and σ tend to zero. The boundary arcs and the coast shrink to zero duration, the coast mass approaches

$$M \rightarrow \frac{2m_i m_{\text{dry}}}{m_i + m_{\text{dry}}}, \quad (105)$$

and the affine uncapped solution of Sec. III is recovered. For finite V , a nonzero central coast is unavoidable under the fixed-power/on-off model. For sufficiently restrictive ceilings, the correct regular structure is not a pair of uniformly scaled symmetric burns separated by a coast. The accelerating and braking branches are generally unequal, and the terminal state

may lie on either a braking interior arc or a maximum-exhaust-speed boundary arc. A formal coast/boundary sliding continuation can appear in the convexified first-order equations, but the exchange argument above excludes it from the true gravity-free minimum-time solution.

The derivation above characterizes normal PMP candidates and eliminates the relaxed sliding candidate by a finite improvement argument. Because the admissible control set (36) is nonconvex, pointwise maximization still supplies only necessary conditions and does not by itself prove global optimality among the remaining regular topologies. It does, however, determine their ordering and reduce each candidate to closed-form arc propagation and a small system of endpoint equations.

The heliocentric model developed next retains the uncapped variable- I_{sp} control law unless a ceiling is imposed explicitly. With a finite ceiling, the pointwise switching logic still applies to the magnitude of the vector primer: an interior thrust arc is followed by a maximum-exhaust-speed boundary arc when the unconstrained thrust magnitude becomes infeasible, and by a coast when the boundary-arc Hamiltonian gain becomes nonpositive. The coast/boundary exchange proof used here relies on gravity-free, position-independent dynamics and therefore does not automatically extend to the heliocentric problem.

V. FULL PLANAR SUN-FIELD MODEL

We now reinstate solar gravity ($\mu = GM_\odot$) and adopt polar coordinates $\mathbf{x} = (r, \theta)$ in the ecliptic. The control is the *planar thrust acceleration* $\mathbf{a}_T = (a_r, a_\theta)$ produced by the variable- I_{sp} engine. The state vector and its dynamics are:

$$\dot{r} = v_r, \quad (106a)$$

$$\dot{\theta} = \frac{v_\theta}{r}, \quad (106b)$$

$$\dot{v}_r = \frac{v_\theta^2}{r} - \frac{\mu}{r^2} + a_r, \quad (106c)$$

$$\dot{v}_\theta = -\frac{v_r v_\theta}{r} + a_\theta, \quad (106d)$$

$$\dot{m} = -\frac{m^2}{2P} (a_r^2 + a_\theta^2). \quad (106e)$$

A. Pontryagin Maximum Principle (PMP)

We again apply the variational framework from Section III-A. For this time-optimal problem, the performance index is $J = \int_0^{t_f} dt$, meaning the Lagrangian is $L = 1$. The components for the Hamiltonian construction are now:

- **State vector:** $\mathbf{x} = (r, \theta, v_r, v_\theta, m)^\top$.
- **Control vector:** $\mathbf{u} = \mathbf{a}_T = (a_r, a_\theta)^\top$.
- **Costate vector:** $\boldsymbol{\lambda} = (\lambda_r, \lambda_\theta, \lambda_{v_r}, \lambda_{v_\theta}, \lambda_m)^\top$.

Using the maximization convention from (3), the Hamiltonian is

$$\mathcal{H} = \boldsymbol{\lambda}^\top \dot{\mathbf{x}} - 1,$$

and it must be maximized by the optimal control at all times. The dot product term $\boldsymbol{\lambda}^\top \dot{\mathbf{x}}$ is constructed by multiplying each

costate by its corresponding state derivative from (106):

$$\begin{aligned} \lambda^\top \dot{\mathbf{x}} = & \lambda_r v_r + \lambda_\theta \frac{v_\theta}{r} + \lambda_{v_r} \left(\frac{v_\theta^2}{r} - \frac{\mu}{r^2} + a_r \right) \\ & + \lambda_{v_\theta} \left(-\frac{v_r v_\theta}{r} + a_\theta \right) - \lambda_m \frac{m^2}{2P} (a_r^2 + a_\theta^2). \end{aligned} \quad (107)$$

Subtracting the running cost $L = 1$ gives the complete Hamiltonian for the Sun-field model. For a minimum-time problem, the PMP asserts that along an optimal trajectory:

- 1) The state and costates evolve according to Hamilton's canonical equations: $\dot{\mathbf{x}} = \partial\mathcal{H}/\partial\lambda$ and $\dot{\lambda} = -\partial\mathcal{H}/\partial\mathbf{x}$.
- 2) The optimal control $\mathbf{a}_T(t)$ maximizes $\mathcal{H}$ point-wise.
- 3) The Hamiltonian remains constant and equal to zero: $\mathcal{H} \equiv 0$.

B. Optimal Control Law

To find the control that maximizes $\mathcal{H}$, we set the partial derivatives of the Hamiltonian with respect to the control variables (a_r, a_θ) to zero:

$$\frac{\partial\mathcal{H}}{\partial a_r} = \lambda_{v_r} - \lambda_m \frac{m^2}{P} a_r = 0, \quad (108a)$$

$$\frac{\partial\mathcal{H}}{\partial a_\theta} = \lambda_{v_\theta} - \lambda_m \frac{m^2}{P} a_\theta = 0. \quad (108b)$$

This yields the optimal thrust acceleration vector:

$$\mathbf{a}_T = k(t) \lambda_v, \quad \text{where} \quad \lambda_v \equiv (\lambda_{v_r}, \lambda_{v_\theta})^\top, \quad (109)$$

and the time-varying gain factor is defined as

$$k(t) \equiv \frac{P}{\lambda_m(t) m^2(t)}. \quad (110)$$

The optimal thrust direction is parallel to the velocity costate vector λ_v . We now show that the gain factor $k(t)$ is, in fact, a constant of motion. Recent adaptive algorithms for low-thrust trajectory design exploit similar primer-vector structures to generate high-quality initial guesses for complex transfers [2].

C. Costate Equations and First Integrals

The costate dynamics are found from $\dot{\lambda}_i = -\partial\mathcal{H}/\partial x_i$:

$$\dot{\lambda}_r = \frac{\lambda_\theta v_\theta}{r^2} + \frac{\lambda_{v_r} v_\theta^2}{r^2} - 2\lambda_{v_r} \frac{\mu}{r^3} - \lambda_{v_\theta} \frac{v_r v_\theta}{r^2}, \quad (111a)$$

$$\dot{\lambda}_\theta = -\frac{\partial\mathcal{H}}{\partial\theta} = 0, \quad (111b)$$

$$\dot{\lambda}_{v_r} = -\lambda_r + \lambda_{v_\theta} \frac{v_\theta}{r}, \quad (111c)$$

$$\dot{\lambda}_{v_\theta} = -\frac{\lambda_\theta}{r} - 2\lambda_{v_r} \frac{v_\theta}{r} + \lambda_{v_\theta} \frac{v_r}{r}, \quad (111d)$$

$$\dot{\lambda}_m = -\frac{\partial\mathcal{H}}{\partial m} = \lambda_m \frac{m}{P} (a_r^2 + a_\theta^2). \quad (111e)$$

From these dynamics, we extract two first integrals.

a) Rotational Symmetry: Equation (111b) arises because the Hamiltonian is independent of the absolute angle θ (a cyclic coordinate). This gives our first integral:

$$\lambda_\theta(t) = C_\theta = \text{const}. \quad (112)$$

b) Mass–Costate Relationship: We find a direct relationship between mass m and its costate λ_m by examining their rates of change. Using the chain rule, we have $\frac{d\lambda_m}{dm} = \frac{\lambda_m}{m}$. Substituting from (111e) and (106):

$$\frac{d\lambda_m}{dm} = \frac{\lambda_m \frac{m}{P} (a_r^2 + a_\theta^2)}{-\frac{m^2}{2P} (a_r^2 + a_\theta^2)} = \frac{\lambda_m m}{-m^2/2} = -2 \frac{\lambda_m}{m}.$$

This is a separable first-order ordinary differential equation:

$$\frac{d\lambda_m}{\lambda_m} = -2 \frac{dm}{m}.$$

Integrating both sides yields $\ln(\lambda_m) = -2 \ln(m) + \text{const}$, which upon exponentiation gives our second, crucial integral of motion:

$$\boxed{\lambda_m(t) m^2(t) = C = \text{const}}. \quad (113)$$

This relation holds for the entire trajectory.

c) Constant Thrust Gain: Substituting the integral from (113) back into the definition of the gain factor (110), we find that it is constant:

$$k(t) = \frac{P}{\lambda_m(t) m^2(t)} = \frac{P}{C} \equiv k_T = \text{const}. \quad (114)$$

The optimal control law (109) therefore simplifies to:

$$\mathbf{a}_T(t) = k_T \lambda_v(t). \quad (115)$$

The physically relevant normal branch has $C \equiv \lambda_m m^2 > 0$, so $k_T = P/C > 0$ and the Hamiltonian is concave in the thrust-acceleration components, confirming that the stationary control maximizes $\mathcal{H}$. The thrust magnitude is proportional to $\|\lambda_v\|$, and the constant gain k_T is determined by the boundary conditions—a key result that significantly simplifies the nature of the optimal control.

This completes the analytical formulation. After using the two first integrals, the orbital and costate shooting dynamics are governed by seven differential equations: the four kinematic/dynamic variables $(r, \theta, v_r, v_\theta)$ and the three non-constant costates $(\lambda_r, \lambda_{v_r}, \lambda_{v_\theta})$. The mass need not be integrated as an independent shooting state, because it can be recovered from the accumulated inverse-mass expenditure,

$$\frac{1}{m(t)} - \frac{1}{m_i} = \int_0^t \frac{a_r(\tau)^2 + a_\theta(\tau)^2}{2P} d\tau. \quad (116)$$

The terminal dry-mass condition is therefore enforced as the scalar quadrature constraint

$$\int_0^{t_f} \frac{a_r(t)^2 + a_\theta(t)^2}{2P} dt = \Delta, \quad \Delta = \frac{1}{m_{\text{dry}}} - \frac{1}{m_i}. \quad (117)$$

A shooting method must find the initial values of the costates, the constant C_θ , the thrust-gain constant k_T , and the final time t_f so that the final circular-orbit boundary conditions and the mass-budget constraint are satisfied.

VI. SIMILARITY REDUCTION OF THE SUN-FIELD TRANSFER FAMILY

The Sun-field boundary-value problem from Sec. V is written in dimensional variables, but its family of circular-to-circular transfers possesses a useful scaling symmetry. This symmetry allows the six nominal inputs

$$(r_0, r_f, \Theta, P, m_i, m_{\text{dry}})$$

to be reduced to three dimensionless parameters for the ideal uncapped model. If a finite exhaust-speed ceiling $v_{e,\text{max}}$ is imposed, two additional dimensionless parameters enter: the normalized ceiling $\nu_{\text{max}} \equiv v_{e,\text{max}}/\sqrt{\mu/r_0}$ and the initial-to-dry mass ratio $\eta \equiv m_i/m_{\text{dry}}$, because the ceiling condition $v_e = 2P/(m\|\mathbf{a}_T\|)$ mixes mass with the other propulsion quantities in a way that κ alone cannot absorb. The capped model therefore depends on $(\rho, \kappa, \Theta, \nu_{\text{max}}, \eta)$ rather than (ρ, κ, Θ) .

The reduction is useful both conceptually and numerically: conceptually, because it shows which combinations of the physical inputs actually determine the transfer shape; numerically, because it makes it possible to tabulate a reduced atlas of normalized solutions and then map them back to dimensional trajectories for any specific spacecraft.

A. Canonical units and normalized variables

Let the departure radius define the distance unit,

$$R_0 \equiv r_0, \quad (118)$$

and let the associated Keplerian time and velocity scales be

$$T_0 \equiv \sqrt{\frac{R_0^3}{\mu}}, \quad V_0 \equiv \frac{R_0}{T_0} = \sqrt{\frac{\mu}{R_0}}. \quad (119)$$

The corresponding acceleration scale is

$$A_0 \equiv \frac{V_0}{T_0} = \frac{\mu}{R_0^2}. \quad (120)$$

We now define the dimensionless independent variable and normalized state by

$$\tau \equiv \frac{t}{T_0}, \quad \bar{r} \equiv \frac{r}{R_0}, \quad \bar{v}_r \equiv \frac{v_r}{V_0}, \quad \bar{v}_\theta \equiv \frac{v_\theta}{V_0}, \quad (121)$$

and similarly normalize the thrust acceleration components as

$$\bar{a}_r \equiv \frac{a_r}{A_0}, \quad \bar{a}_\theta \equiv \frac{a_\theta}{A_0}. \quad (122)$$

The polar angle θ is already dimensionless and is left unchanged.

The mass is handled more conveniently through the inverse-mass increment that already appears in the free-space solution. Define

$$\Delta \equiv \frac{1}{m_{\text{dry}}} - \frac{1}{m_i}, \quad (123)$$

and introduce the normalized fuel coordinate

$$q(\tau) \equiv \frac{m(\tau)^{-1} - m_i^{-1}}{\Delta}. \quad (124)$$

By construction,

$$q(0) = 0, \quad q(\tau_f) = 1, \quad (125)$$

so q measures the fraction of the available inverse-mass budget that has been consumed. The dimensional mass is recovered from q as

$$m(\tau) = \frac{1}{m_i^{-1} + \Delta q(\tau)}. \quad (126)$$

B. Dimensionless form of the dynamics

Substituting the scaled variables into (106) gives

$$\bar{r}' = \bar{v}_r, \quad (127a)$$

$$\theta' = \frac{\bar{v}_\theta}{\bar{r}}, \quad (127b)$$

$$\bar{v}_r' = \frac{\bar{v}_\theta^2}{\bar{r}} - \frac{1}{\bar{r}^2} + \bar{a}_r, \quad (127c)$$

$$\bar{v}_\theta' = -\frac{\bar{v}_r \bar{v}_\theta}{\bar{r}} + \bar{a}_\theta, \quad (127d)$$

$$q' = \frac{\bar{a}_r^2 + \bar{a}_\theta^2}{\kappa}, \quad (127e)$$

where prime denotes differentiation with respect to τ , and where the single dimensionless propulsion parameter is

$$\kappa \equiv 2P\Delta \frac{R_0^{5/2}}{\mu^{3/2}} = 2P \left(\frac{1}{m_{\text{dry}}} - \frac{1}{m_i} \right) \frac{r_0^{5/2}}{\mu^{3/2}}. \quad (128)$$

This is the key reduction. In dimensional form, the mass-flow law (2) depends separately on P , m_i , m_{dry} , and r_0 . After scaling, those four quantities appear only through the single combination κ . The normalized equations therefore no longer distinguish between spacecraft that have different powers and masses but the same value of κ . Such spacecraft follow the same dimensionless trajectory and differ only by the dimensional scales used to reconstruct time, radius, velocity, and acceleration.

C. Dimensionless boundary conditions

For a transfer from a circular orbit of radius r_0 to a circular orbit of radius r_f , the normalized boundary conditions become

$$\bar{r}(0) = 1, \quad \bar{r}(\tau_f) = \rho, \quad (129a)$$

$$\theta(0) = 0, \quad \theta(\tau_f) = \Theta, \quad (129b)$$

$$\bar{v}_r(0) = 0, \quad \bar{v}_r(\tau_f) = 0, \quad (129c)$$

$$\bar{v}_\theta(0) = 1, \quad \bar{v}_\theta(\tau_f) = \rho^{-1/2}, \quad (129d)$$

$$q(0) = 0, \quad q(\tau_f) = 1, \quad (129e)$$

with

$$\rho \equiv \frac{r_f}{r_0}. \quad (130)$$

The quantity ρ is the radius ratio, while

$$\Theta \equiv \theta_f - \theta_0 + 2\pi N \quad (131)$$

is the unwrapped transfer angle, with $N \in \mathbb{Z}_{\geq 0}$ selecting the number of completed revolutions. Only the angular difference matters, not the absolute departure angle, because the Sun-field model is rotationally invariant. This is the same symmetry that produced the first integral $\lambda_\theta = C_\theta$ in Sec. V.

Equations (127)–(129) show that, for the uncapped variable- I_{sp} model, the normalized transfer problem depends only on

$$(\rho, \kappa, \Theta).$$

This is the reduced parameter space for the ideal uncapped model.

D. Why the parameter space shrinks

It is useful to state explicitly what has been removed.

First, the absolute orbital scale r_0 disappears from the normalized equations because lengths are measured in units of r_0 , times in units of $\sqrt{r_0^3/\mu}$, and velocities in units of $\sqrt{\mu/r_0}$. Only the ratio r_f/r_0 remains.

Second, the three propulsion quantities P , m_i , and m_{dry} do not independently affect the normalized dynamics. They enter only through the inverse-mass budget Δ and then only through the combination κ in (128). Thus three nominal propulsion parameters collapse to one similarity parameter.

Third, the absolute phase angle θ_0 is irrelevant because the equations are invariant under a rigid rotation of the entire trajectory. Only the total angular change Θ matters.

As a result, the circular-to-circular Sun-field transfer problem is parameterized, at the similarity level, by the three variables (ρ, κ, Θ) rather than the six nominal quantities $(r_0, r_f, \Theta, P, m_i, m_{\text{dry}})$.

E. Forward map from nominal parameters to similarity space

For a given dimensional transfer request, the reduced coordinates are obtained directly from the nominal mission parameters:

$$\rho = \frac{r_f}{r_0}, \quad (132)$$

$$\Theta = \theta_f - \theta_0 + 2\pi N, \quad (133)$$

$$\Delta = \frac{1}{m_{\text{dry}}} - \frac{1}{m_i}, \quad (134)$$

$$\kappa = 2P\Delta \frac{r_0^{5/2}}{\mu^{3/2}}. \quad (135)$$

Here N selects the desired multi-revolution branch. Thus the similarity reduction is applied by first computing (ρ, κ, Θ) from the physical problem and then solving the corresponding normalized boundary-value problem.

F. Inverse map from a normalized atlas solution to dimensional variables

Suppose now that a normalized solution has been computed or interpolated in the reduced space (ρ, κ, Θ) . That solution consists of the dimensionless functions

$$\bar{r}(\tau), \quad \theta(\tau), \quad \bar{v}_r(\tau), \quad \bar{v}_\theta(\tau), \quad q(\tau), \quad \bar{a}_r(\tau), \quad \bar{a}_\theta(\tau),$$

together with the normalized final time τ_f .

For the specific dimensional system defined by r_0 , m_i , m_{dry} , and μ , the trajectory is recovered by the inverse scaling

$$t = T_0\tau = \sqrt{\frac{r_0^3}{\mu}} \tau, \quad (136)$$

$$r(t) = r_0 \bar{r}(\tau), \quad (137)$$

$$\theta(t) = \theta_0 + \theta(\tau), \quad (138)$$

$$v_r(t) = \sqrt{\frac{\mu}{r_0}} \bar{v}_r(\tau), \quad (139)$$

$$v_\theta(t) = \sqrt{\frac{\mu}{r_0}} \bar{v}_\theta(\tau), \quad (140)$$

$$a_r(t) = \frac{\mu}{r_0^2} \bar{a}_r(\tau), \quad (141)$$

$$a_\theta(t) = \frac{\mu}{r_0^2} \bar{a}_\theta(\tau), \quad (142)$$

$$m(t) = \frac{1}{m_i^{-1} + \Delta q(\tau)}. \quad (143)$$

In particular, the dimensional flight time is

$$t_f = \sqrt{\frac{r_0^3}{\mu}} \tau_f. \quad (144)$$

This gives the practical recipe for using a precomputed solution atlas:

- 1) Start from the nominal physical parameters.
- 2) Compute ρ , Θ , and κ from the forward map above.
- 3) Retrieve or interpolate the normalized solution stored at that location in (ρ, κ, Θ) -space.
- 4) Recover the dimensional trajectory by the inverse scaling formulas.

G. What should be stored in the atlas

For numerical continuation, it is not necessary to store the full dimensional trajectory. A more efficient choice is to store the normalized shooting data, namely the initial costates, the constant C_θ associated with rotational symmetry, the thrust-gain constant k_T , and the normalized flight time. For the uncapped model, because these quantities are stored in dimensionless form, the same atlas entry can be reused for any dimensional spacecraft that maps to the same (ρ, κ, Θ) . If a finite exhaust-speed ceiling is included, the atlas must also be indexed by ν_{max} and η . This is the computational advantage of the similarity reduction: one normalized solution represents an entire equivalence class of physical missions.

VII. NUMERICAL PROCEDURE

Solving the two-point boundary-value problem and constructing PMP-extremal transfer candidates requires a numerical approach. We employ a two-stage strategy that combines a global search to find an initial, feasible trajectory with a local refinement process that uses a continuation method to precisely reach the desired target [3], [11].

A. Objective Function

The core of the numerical method is an objective function, J , which quantifies endpoint targeting accuracy and constraint satisfaction for a candidate trajectory. This function is minimized to find a set of initial costate parameters that satisfies the endpoint and constraint residuals as accurately as possible. It incorporates penalties for violating physical constraints and errors for missing the target state. The function has two distinct forms depending on the optimization stage.

1) *Global Search Objective*: In the initial search phase, the goal is to find *any* feasible trajectory whose terminal state lies on some circular heliocentric orbit beyond a minimum radius, without fixing the specific final radius or polar angle. The target radius and final polar angle are pinned only in Stage 2. The Stage 1 objective function J_{global} therefore rewards reaching the local circular velocity at whatever final radius the trajectory attains in the outer solar system, while penalizing physically unrealistic outcomes:

$$J_{\text{global}} = w_f \mathcal{P}_f + w_r \mathcal{P}_r + w_v \mathcal{E}_v^2. \quad (145)$$

The terms are defined as:

- $\mathcal{P}_f$: A fuel penalty that applies a large constant value if the final mass m_f drops below the dry mass m_{dry} , effectively creating a hard constraint against using more fuel than available.
- $\mathcal{P}_r$: A radial position penalty, $\max(0, r_{\min} - r_f)^2$, which discourages solutions that fail to travel a minimum distance $r_{\min}$ (e.g., 1.5 AU) from the Sun.
- $\mathcal{E}_v$: The velocity mismatch error at the final state. It measures the Euclidean distance between the final velocity vector $\mathbf{v}_f = (v_{r,f}, v_{\theta,f})$ and the required velocity for a circular orbit at the final radius r_f , given by $\mathbf{v}_{\text{circ}} = (0, \sqrt{\mu/r_f})$. The squared error is $\mathcal{E}_v^2 = v_{r,f}^2 + (v_{\theta,f} - \sqrt{\mu/r_f})^2$.
- w_f, w_r, w_v : Positive weighting constants that balance the contribution of each term.

2) *Local Refinement Objective*: Once an initial solution is found, the objective function is switched to a targeted version, J_{local} , to steer the trajectory to a precise destination $(r_{\text{des}}, \theta_{\text{des}})$:

$$J_{\text{local}} = w_f \mathcal{P}_f + w_v \mathcal{E}_v^2 + w_r (r_f - r_{\text{des}})^2 + w_\theta (\theta_f - \theta_{\text{des}})^2. \quad (146)$$

Here, the radial penalty $\mathcal{P}_r$ is removed, and two new terms are added to penalize deviations from the desired final radius r_{des} and polar angle θ_{des} .

B. Two-Stage Optimization Strategy

1) *Stage 1: Global Search*: An initial, admissible solution is found by minimizing the global objective function (145). For this task, we use the `differential_evolution` algorithm from the SciPy library [14]. This robust, population-based method is well-suited for exploring the wide parameter space of the initial costates without getting trapped in local minima, thereby finding a good starting point for the next stage.

2) *Stage 2: Continuation and Local Refinement*: Starting from the solution found in Stage 1, which ends at a point (r^*, θ^*) , we use a continuation (or homotopy) method to incrementally approach the final desired target $(r_{\text{des}}, \theta_{\text{des}})$ [15]. The procedure is as follows:

- 1) Set an intermediate target partway between the current known solution's endpoint and the final desired endpoint.
- 2) Perform a local optimization by minimizing the targeted objective function (146) using Powell's method [16] (derivative-free) as implemented in `scipy.optimize.minimize`. The initial guess for the optimizer is the set of parameters from the previous successfully converged solution.
- 3) If the optimizer converges to a high-precision solution, the new solution is accepted. The intermediate target is then advanced closer to the final destination, and the process repeats.
- 4) If the optimizer fails to converge, it indicates the step to the new target was too large. The intermediate target is then moved back, typically halving the distance between the last successful endpoint and the current target, and the optimization is re-attempted.

This adaptive stepping keeps the local optimizer close to a previously converged solution and substantially improves robustness, although convergence is not guaranteed for every target or every continuation step.

3) *Qualification of Time-Optimality Claims*: The indirect formulation enforces the PMP state equations, costate equations, and pointwise stationarity of the Hamiltonian by construction. Therefore, a converged shooting solution is a PMP extremal and is a necessary-condition candidate for a time-optimal transfer.

However, satisfying the PMP conditions alone does not prove global optimality, and by itself it also does not prove local optimality in the strict second-order sense. A rigorous local optimality proof would require additional second-order analysis, such as checking for conjugate points, or an equivalent sufficiency test. In this work we therefore describe the numerical trajectories as time-optimal candidates or PMP-extremal transfer candidates. The reported times should be interpreted as the transfer times of the converged extremals found by the continuation procedure, not as a proof that no shorter feasible transfer exists.

All simulations, optimization procedures, and figures were generated using the accompanying open-source Python implementation [17]. The repository includes the complete solver, plotting scripts, and resulting graphs of trajectories, fuel and acceleration magnitudes.

VIII. RESULTS

The numerical procedure was applied to compute PMP-extremal transfer candidates from a 1 AU circular orbit (representing Earth) to various target circular orbits. The targets are defined by their final radius and the heliocentric polar angle $\theta_f - \theta_0$ relative to the starting position. (We retain the symbol θ for this polar angle throughout; it is not the orbital

true anomaly, which is undefined on a circular orbit.) The propulsion model used $P = 1$ GW of useful jet power, with an initial vehicle mass of 3000 t and a dry mass of 1000 t. As an order-of-magnitude check on these numbers, the free-space benchmark of Sec. III with $L = r_f - r_0 = 4.2$ AU, $P = 1$ GW, and $\Delta = 1/m_{\text{dry}} - 1/m_i$ gives

$$T_{\min} = \left[\frac{6L^2}{P\Delta} \right]^{1/3} \approx 177 \text{ days}.$$

The Sun-field transfers to 5.2 AU in Tables I–II range from about 170 to 215 days and bracket this value. The flight times shorter than the free-space benchmark are possible because the spacecraft begins with a non-zero tangential velocity (~ 30 km/s at 1 AU) and can convert solar gravitational potential energy along the transfer. The longer ones reflect the additional arc length traversed around the Sun, the energy required to climb the potential well to the outer orbit, and, in some cases, the cost of the low-perihelion Oberth pass.

Figures 1 and 2 show the relationship between flight time and the orbital geometry. The flight times exhibit a U-shaped dependence on the final polar angle, with a clear minimum for transfers that are neither direct short-arcs nor long-arcs. Similar behavior has been reported in recent multi-objective studies of variable- I_{sp} maneuvers [11].

Figures 3 and 5 show example candidate extremals. A key feature of these trajectories is the inclusion of an Oberth-like low-perihelion maneuver. To gain orbital energy efficiently, the spacecraft first thrusts to lower its perihelion, making a low-perihelion pass around the Sun. It then applies its main acceleration burn at maximum velocity near the Sun, which is far more effective at increasing orbital energy than thrusting in a higher orbit [18]. This is visible in the plots as an initial dip towards the Sun and a corresponding peak in the acceleration magnitude. These Oberth-type solutions should be interpreted within the idealized dynamical model used here. No explicit constraints are imposed on minimum perihelion distance, solar heating, radiator performance near the Sun, pointing restrictions, or power-system survivability. Adding such constraints would generally increase the transfer times or remove some of the most aggressive solar-pass solutions.

Table I and Table II present the resulting transfer times in days for a grid of targets. As expected, transfers to more distant orbits take longer. The notation “N/A” indicates that the numerical continuation and shooting procedure did not find a trajectory satisfying the terminal circular-orbit constraints before violating the dry-mass bound. Such cases may correspond to a genuine feasibility boundary for the specified power and propellant budget, or to a numerical convergence limitation of the present solver. Figure 5 shows a trajectory close to such a boundary, with an extreme solar pass that cannot be extended further within the imposed model constraints.

IX. CONCLUSION

We combined analytic insight (two closed-form first integrals and a free-space benchmark solution) with a continuation strategy for constructing time-optimal transfer candidates for a constant-power, variable- I_{sp} engine. The method successfully

generates PMP-extremal candidate trajectories, including those featuring low-perihelion Oberth-like maneuvers, for transfers out to the giant planets with a gigawatt-class useful jet-power engine. Recent developments in variable- I_{sp} propulsion and adaptive low-thrust guidance algorithms [2], [11] suggest that further improvements in mission performance are possible. Future work will extend this framework to three-dimensional rendezvous problems. A rigorous demonstration of local or global time optimality for the Sun-field examples remains future work; it would require either second-order sufficiency analysis of the extremals, such as conjugate-point checks, or a broader global comparison across competing transfer branches.

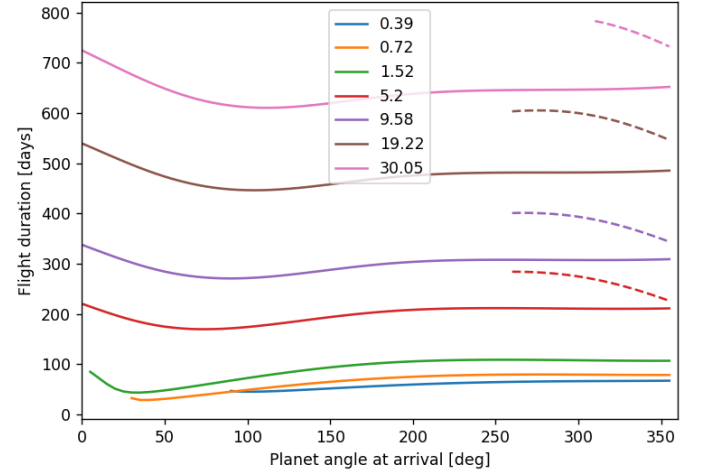


Fig. 1. Flight times in days versus final polar angle at arrival.

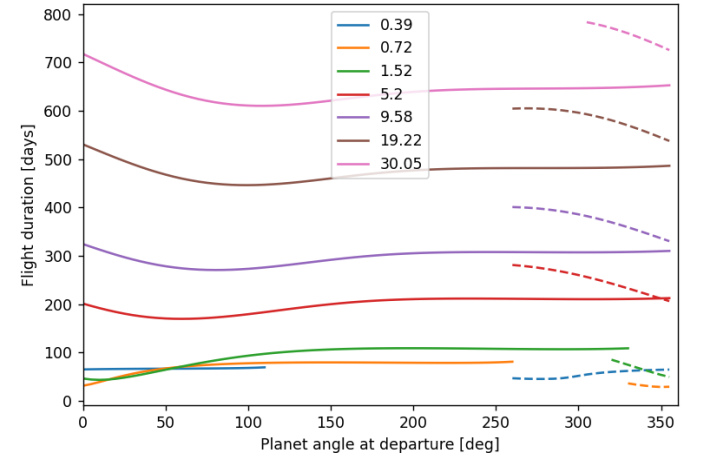


Fig. 2. Flight times in days versus initial polar angle at departure.

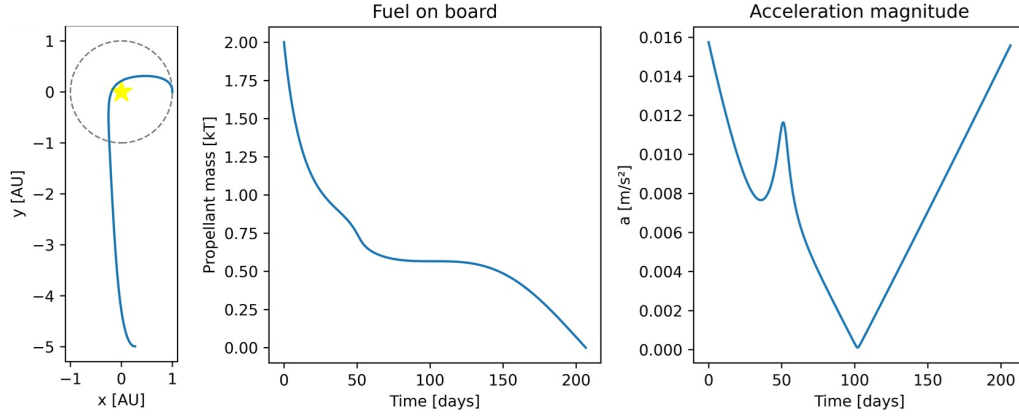


Fig. 3. Example trajectory from $r = 1$ AU, $\theta = 0^\circ$ to $r = 5$ AU, $\theta = 273.1^\circ$ in 207 days. The trajectory includes a low-perihelion Oberth-like pass to gain orbital energy, which is apparent from the acceleration peak around day 50.

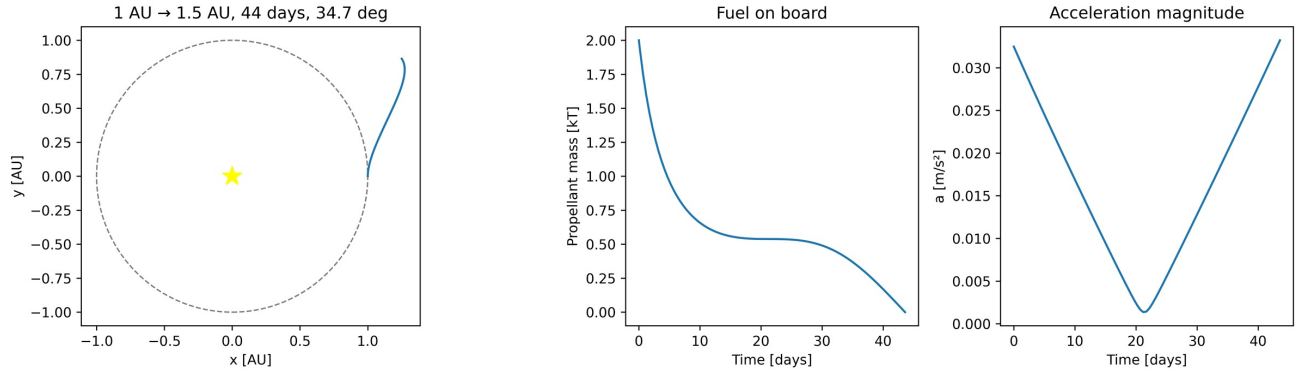


Fig. 4. Example trajectory from $r = 1$ AU, $\theta = 0^\circ$ to $r = 1.5$ AU, $\theta = 34.7^\circ$ in 44 days. The trajectory is a very close analogy to the trajectories without gravity, demonstrating clear acceleration-deceleration pattern that is almost fully symmetric.

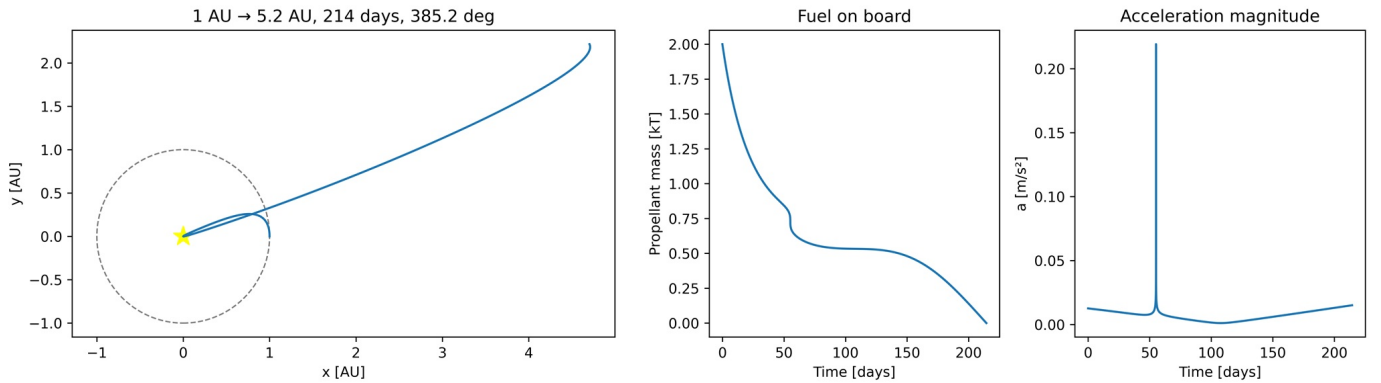


Fig. 5. An example of a more extreme low-perihelion Oberth-like maneuver for a transfer to Jupiter's orbit. The trajectory is from $r = 1$ AU, $\theta = 0^\circ$ to $r = 5.2$ AU, $\theta = 385.2^\circ$ and lasts 214 days.

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APPENDIX A

POWER-SYSTEM MASS CLOSURE FOR THE FREE-SPACE BENCHMARK

The benchmark result (30) was derived for prescribed initial and dry masses. For high-power spacecraft, however, the dry mass may itself depend on the propulsion system, because both the power-conversion hardware and the radiators must be carried together with the payload. This appendix shows how the free-space minimum-time law is modified under a simple power-system mass closure.

Let P denote the useful jet power appearing in the rocket model, and let P_{tot} be the total onboard engine power. Define

$$\alpha_{\text{eng}} \equiv \frac{P_{\text{tot}}}{m_{\text{eng}}}, \quad \phi \equiv \frac{P_{\text{heat}}}{P_{\text{tot}}}, \quad \rho_{\text{rad}} \equiv \frac{P_{\text{heat}}}{m_{\text{rad}}}, \quad (147)$$

where α_{eng} is the engine specific power, ϕ is the waste-heat fraction, and ρ_{rad} is the radiator specific heat-rejection capability. Since only the fraction $1 - \phi$ of the total power is converted into useful jet power,

$$P = (1 - \phi)P_{\text{tot}}, \quad P_{\text{heat}} = \phi P_{\text{tot}} = \frac{\phi}{1 - \phi}P. \quad (148)$$

Therefore,

$$m_{\text{eng}} = \frac{P}{(1 - \phi)\alpha_{\text{eng}}}, \quad m_{\text{rad}} = \frac{\phi}{1 - \phi} \frac{P}{\rho_{\text{rad}}}. \quad (149)$$

Let m_{pay} be the payload mass. The dry mass can then be written in the affine form

$$m_{\text{dry}} = m_{\text{pay}} + m_{\text{eng}} + m_{\text{rad}} = m_{\text{pay}} + \beta P, \quad (150)$$

with

$$\beta \equiv \frac{1}{(1 - \phi)\alpha_{\text{eng}}} + \frac{\phi}{(1 - \phi)\rho_{\text{rad}}}. \quad (151)$$

The quantity β has units of kg/W and is the inverse of the effective propulsion-system specific power,

$$\alpha_{\text{sys}} \equiv \beta^{-1}. \quad (152)$$

To close the mass model, assume a fixed fuel-to-dry-mass ratio

$$\mathcal{R} \equiv \frac{m_{\text{fuel}}}{m_{\text{dry}}}. \quad (153)$$

Then

$$m_i = (1 + \mathcal{R})m_{\text{dry}}, \quad (154)$$

and the parameter Δ from (30) becomes

$$\Delta = \frac{1}{m_{\text{dry}}} - \frac{1}{m_i} = \frac{\mathcal{R}}{(1 + \mathcal{R})m_{\text{dry}}}. \quad (155)$$

Substituting (150) and (155) into the benchmark law (30) yields

$$T(P) = \left[\frac{6L^2(1 + \mathcal{R})}{\mathcal{R}} \left(\frac{m_{\text{pay}}}{P} + \beta \right) \right]^{1/3}. \quad (156)$$

Equation (156) is the free-space minimum-time law when the dry mass is allowed to scale linearly with propulsion power.

Several immediate consequences follow. First,

$$T^3(P) = \frac{6L^2(1 + \mathcal{R})}{\mathcal{R}} \left(\frac{m_{\text{pay}}}{P} + \beta \right), \quad (157)$$

so that

$$\frac{d}{dP}T^3(P) = -\frac{6L^2(1 + \mathcal{R})}{\mathcal{R}} \frac{m_{\text{pay}}}{P^2} < 0. \quad (158)$$

Under the idealized assumptions of this closure—unlimited exhaust speed, fixed waste-heat fraction ϕ , fixed engine specific power α_{eng} , and fixed radiator specific heat rejection ρ_{rad} , all independent of operating point—the transfer time decreases monotonically with power and has no finite mathematical optimum:

$$P^* \rightarrow \infty. \quad (159)$$

This formal limit should not be interpreted as a physically achievable design target, since at very large P the assumed linear mass closure breaks down: in practice ρ_{rad} depends on the radiator operating temperature, α_{eng} is bounded by materials and shielding, and the exhaust speed itself is bounded above. The relevant practical implication is rather the second observation: the family approaches the asymptotic lower bound

$$T_{\infty} = \left[\frac{6L^2(1 + \mathcal{R})}{\mathcal{R}} \beta \right]^{1/3}. \quad (160)$$

This lower bound arises because increasing P also increases the dry mass that must be accelerated.

A convenient engineering scale is obtained by equating propulsion-hardware mass to payload mass,

$$m_{\text{eng}} + m_{\text{rad}} = m_{\text{pay}}. \quad (161)$$

Using (150), this defines the characteristic power

$$\beta P_{\text{knee}} = m_{\text{pay}}, \quad P_{\text{knee}} = \frac{m_{\text{pay}}}{\beta} = \alpha_{\text{sys}} m_{\text{pay}}. \quad (162)$$

At this point the dry mass is $m_{\text{dry}} = 2m_{\text{pay}}$, and (156) can be written as

$$T(P) = T_{\infty} \left(1 + \frac{P_{\text{knee}}}{P} \right)^{1/3}. \quad (163)$$

Hence

$$T(P_{\text{knee}}) = 2^{1/3}T_{\infty}. \quad (164)$$

Equations (156)–(164) provide an engineering interpretation of the free-space benchmark. They are not used in the subsequent Sun-field optimal-control formulation, but they show how the cubic-root time law is modified when propulsion-system mass scales with power.

APPENDIX B

DIRECT METHOD USING SPLINE PARAMETERIZATION

Before adopting the indirect method based on Pontryagin's Maximum Principle (PMP) detailed in the main body of this paper, a more straightforward *direct method* was explored. Direct methods seek to find the optimal trajectory by parameterizing the path with a set of functions and then using a nonlinear programming solver to find the best parameters. This approach avoids the complexity of deriving and solving the costate equations.

Our implementation parameterized the spacecraft's trajectory, $\mathbf{r}(t) = (x(t), y(t))$, using cubic splines. The core idea is to define the path by a small number of control points, which become the variables for optimization.

The procedure was as follows:

1. A fixed number of N control points, $(\mathbf{p}_i, t_i)$ for $i = 0, \dots, N-1$, were defined along the time axis of the transfer. The endpoints, $\mathbf{p}_0$ and $\mathbf{p}_{N-1}$, were fixed to match the start and end positions of the transfer arc. The initial and final velocities were enforced by connecting the transfer arc to the initial and final circular orbits.
2. The positions of the intermediate $N-2$ control points were the free variables for the optimization.
3. For any given set of control points, two cubic splines, $x_s(t)$ and $y_s(t)$, were fitted to create a continuous and smooth trajectory $\mathbf{r}(t) = (x_s(t), y_s(t))$. A cubic spline is defined piecewise by third-order polynomials, ensuring that the first and second derivatives (velocity and acceleration) are continuous across the entire path.
4. The required velocity $\dot{\mathbf{r}}(t)$ and acceleration $\ddot{\mathbf{r}}(t)$ vectors were obtained by numerically differentiating the spline functions.
5. With the total acceleration $\ddot{\mathbf{r}}(t)$ known, the required thrust acceleration was calculated by subtracting the gravitational component:

$$\mathbf{a}_T(t) = \ddot{\mathbf{r}}(t) - \mathbf{a}_g(\mathbf{r}(t)). \quad (165)$$

6. The instantaneous mass flow rate was then determined from the thrust acceleration magnitude using (2). Integrating this rate backwards in time yielded the total propellant consumed for the maneuver.
7. A nonlinear optimizer (specifically, the Nelder-Mead method) was used to adjust the positions of the intermediate control points to minimize a cost function based on the final fuel expenditure and flight time.

Limitations and Stagnation in Local Minima

While conceptually simple, this direct method proved inadequate for finding high-performance trajectories. The primary difficulty was the nature of the optimization landscape. The relationship between the position of a control point and the final mission cost is highly complex and nonlinear. As a result, the search space is filled with a vast number of local minima.

For a very small number of intermediate control points ($N = 3$ or $N = 4$ total), the method could find a reasonable, smooth trajectory. However, as the number of control points was increased to grant the path more freedom and potential

for higher performance, the dimensionality of the search space grew, and the solver would invariably become trapped in a suboptimal local minimum.

An attempt was made to mitigate this by using a continuation strategy on the number of control points: first, an optimal path with $N = 4$ points was found. Then, a fifth point was added at the location of maximum curvature, and the system was re-optimized. While this yielded marginal improvements, the solution quality consistently stagnated far below that achievable with the PMP-based method. The direct spline method lacks the rigorous guidance provided by the costate (or "primer vector") variables, which point the way toward optimality across the entire trajectory. These difficulties ultimately motivated the adoption of the indirect shooting method presented in this work.

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TABLE I
TRANSFER TIME, IN DAYS, OF THE CONVERGED PMP-EXTREMAL CANDIDATE FROM A CIRCULAR ORBIT OF 1 AU AND FINAL POLAR ANGLE (AT
ROCKET ARRIVAL).

Polar angle (°)	Target Radius (AU)						
	0.390	0.720	1.520	5.200	9.580	19.220	30.050
0	N/A	N/A	N/A	220.50	337.82	539.55	724.65
5	N/A	N/A	84.99	214.66	331.66	532.47	716.76
10	N/A	N/A	73.01	208.91	325.52	525.31	708.75
15	N/A	N/A	60.88	203.32	319.44	518.17	700.69
20	N/A	N/A	51.35	197.98	313.53	511.09	692.64
25	N/A	N/A	45.94	192.95	307.83	504.13	684.71
30	N/A	32.66	43.80	188.35	302.40	497.40	676.93
35	N/A	29.03	43.63	184.23	297.29	490.93	669.37
40	N/A	28.92	44.61	180.59	292.59	484.76	662.08
45	N/A	29.86	46.15	177.47	288.32	478.95	655.19
50	N/A	31.19	48.07	174.94	284.48	473.58	648.68
55	N/A	32.74	50.23	172.94	281.10	468.66	642.60
60	N/A	34.42	52.55	171.44	278.25	464.20	637.00
65	N/A	36.18	54.96	170.42	275.88	460.21	631.92
70	N/A	38.00	57.46	169.90	273.97	456.76	627.37
75	N/A	39.84	59.98	169.80	272.50	453.79	623.38
80	N/A	41.69	62.51	170.05	271.54	451.39	619.93
85	N/A	43.55	65.05	170.65	270.98	449.42	617.06
90	46.97	45.39	67.58	171.57	270.81	447.98	614.72
95	45.78	47.22	70.07	172.77	270.99	447.01	612.89
100	45.47	49.04	72.55	174.17	271.54	446.46	611.58
105	45.54	50.82	74.97	175.77	272.38	446.33	610.73
110	45.86	52.58	77.35	177.53	273.50	446.57	610.37
115	46.37	54.30	79.68	179.42	274.83	447.19	610.43
120	47.00	55.99	81.93	181.41	276.38	448.11	610.89
125	47.72	57.63	84.12	183.48	278.09	449.29	611.70
130	48.49	59.22	86.24	185.58	279.93	450.73	612.82
135	49.31	60.76	88.27	187.70	281.87	452.36	614.21
140	50.16	62.24	90.22	189.81	283.87	454.17	615.83
145	51.02	63.68	92.07	191.90	285.92	456.10	617.64
150	51.89	65.05	93.83	193.93	287.97	458.09	619.57
155	52.75	66.36	95.50	195.89	289.99	460.14	621.61
160	53.61	67.61	97.05	197.76	291.96	462.19	623.69
165	54.46	68.79	98.50	199.53	293.86	464.22	625.79
170	55.28	69.89	99.84	201.18	295.66	466.18	627.86
175	56.08	70.94	101.08	202.72	297.35	468.07	629.88
180	56.86	71.91	102.22	204.12	298.92	469.85	631.81
185	57.61	72.83	103.25	205.38	300.34	471.50	633.63
190	58.34	73.67	104.18	206.51	301.63	473.03	635.31
195	59.03	74.46	105.02	207.53	302.79	474.41	636.87
200	59.68	75.17	105.75	208.39	303.79	475.63	638.27
205	60.31	75.83	106.40	209.13	304.67	476.72	639.53
210	60.90	76.42	106.97	209.77	305.42	477.67	640.64
215	61.45	76.96	107.46	210.31	306.05	478.48	641.61
220	61.97	77.44	107.86	210.74	306.57	479.17	642.45
225	62.47	77.86	108.20	211.07	306.99	479.74	643.16
230	62.93	78.23	108.48	211.33	307.32	480.22	643.76
235	63.35	78.56	108.69	211.52	307.57	480.60	644.27
240	63.74	78.84	108.85	211.65	307.75	480.91	644.68
245	64.10	79.07	108.95	211.72	307.87	481.14	645.02
250	64.44	79.26	109.01	211.75	307.94	481.31	645.29
255	64.74	79.42	109.03	211.73	307.97	481.43	645.51
260	65.02	79.54	109.01	211.68	307.96	481.51	645.68
265	65.26	79.62	108.96	211.60	307.92	481.57	645.82
270	65.48	79.66	108.88	211.50	307.87	481.59	645.94
275	65.69	79.69	108.78	211.40	307.79	481.61	646.04
280	65.86	79.69	108.66	211.27	307.72	481.62	646.12
285	66.02	79.65	108.52	211.14	307.63	481.62	646.22
290	66.16	79.61	108.37	211.00	307.55	481.64	646.31
295	66.27	79.54	108.21	210.88	307.48	481.67	646.43
300	66.38	79.47	108.03	210.75	307.42	481.71	646.57
305	66.47	79.37	107.87	210.65	307.38	481.79	646.74
310	66.56	79.27	107.70	210.56	307.37	481.89	646.96
315	66.63	79.18	107.54	210.49	307.38	482.04	647.22
320	66.70	79.07	107.39	210.45	307.43	482.23	647.54
325	66.77	78.98	107.26	210.43	307.52	482.48	647.92
330	66.84	78.88	107.14	210.45	307.64	482.79	648.38
335	66.90	78.81	107.03	210.51	307.82	483.16	648.91
340	66.99	78.74	106.97	210.61	308.06	483.61	649.54
345	67.08	78.70	106.94	210.75	308.36	484.14	650.27
350	67.19	78.70	106.94	210.96	308.71	484.76	651.11
355	67.32	78.73	107.00	211.22	309.15	485.47	652.06
360	67.49	78.81	107.10	211.54	309.66	486.29	653.13

TABLE II
TRANSFER TIME, IN DAYS, OF THE CONVERGED PMP-EXTREMAL CANDIDATE FROM A CIRCULAR ORBIT OF 1 AU AND INITIAL POLAR ANGLE (AT
ROCKET DEPARTURE).

Polar Angle (°)	Target Radius (AU)						
	0.390	0.720	1.520	5.200	9.580	19.220	30.050
-100	46.96	N/A	N/A	281.19	401.02	604.63	N/A
-95	46.26	N/A	N/A	279.72	400.45	605.12	N/A
-90	45.76	N/A	N/A	277.93	399.55	605.20	N/A
-85	45.48	N/A	N/A	275.79	398.22	604.85	N/A
-80	45.50	N/A	N/A	273.34	396.54	604.07	N/A
-75	45.92	N/A	N/A	270.57	394.49	602.85	N/A
-70	47.06	N/A	N/A	267.50	392.06	601.12	N/A
-65	49.08	N/A	N/A	264.11	389.20	598.91	N/A
-60	51.87	N/A	N/A	260.40	385.96	596.20	N/A
-55	54.65	N/A	N/A	256.39	382.34	592.99	783.40
-50	56.90	N/A	N/A	252.12	378.37	589.31	779.93
-45	58.68	N/A	N/A	247.60	374.05	585.12	775.84
-40	60.08	N/A	85.30	242.86	369.37	580.49	771.21
-35	61.19	N/A	80.06	237.92	364.40	575.39	766.05
-30	62.11	36.38	74.72	232.82	359.17	569.87	760.36
-25	62.88	33.69	69.37	227.58	353.71	564.00	754.17
-20	63.51	31.65	64.02	222.28	348.06	557.81	747.56
-15	64.05	29.77	58.87	216.96	342.25	551.34	740.58
-10	64.50	28.86	53.88	211.67	336.37	544.62	733.26
-5	64.89	29.30	49.74	206.46	330.45	537.74	725.67
0	65.21	31.38	46.38	201.41	324.55	530.74	717.88
5	65.48	34.85	44.28	196.58	318.72	523.70	709.96
10	65.72	39.16	43.59	192.04	313.06	516.69	701.98
15	65.92	43.78	44.17	187.83	307.58	509.73	694.01
20	66.09	48.26	45.84	184.00	302.34	502.93	686.11
25	66.23	52.44	48.21	180.58	297.40	496.34	678.37
30	66.35	56.21	51.07	177.61	292.82	490.00	670.84
35	66.45	59.57	54.25	175.13	288.65	483.96	663.57
40	66.55	62.47	57.60	173.15	284.87	478.29	656.62
45	66.63	65.02	61.04	171.63	281.52	473.04	650.08
50	66.70	67.23	64.51	170.57	278.62	468.22	643.97
55	66.77	69.13	67.95	169.94	276.22	463.83	638.30
60	66.84	70.78	71.31	169.78	274.27	459.92	633.10
65	66.91	72.19	74.60	169.99	272.76	456.56	628.48
70	67.01	73.43	77.78	170.54	271.68	453.63	624.33
75	67.11	74.48	80.81	171.40	271.05	451.26	620.80
80	67.24	75.39	83.70	172.59	270.82	449.33	617.74
85	67.41	76.18	86.43	174.01	270.94	447.93	615.27
90	67.62	76.85	88.98	175.63	271.40	446.97	613.30
95	67.90	77.44	91.37	177.43	272.19	446.44	611.88
100	68.28	77.91	93.55	179.39	273.28	446.33	610.94
105	68.79	78.33	95.58	181.45	274.59	446.59	610.43
110	69.52	78.67	97.41	183.58	276.10	447.23	610.40
115	N/A	78.96	99.09	185.77	277.82	448.16	610.75
120	N/A	79.18	100.57	187.96	279.67	449.36	611.45
125	N/A	79.36	101.92	190.15	281.62	450.81	612.49
130	N/A	79.50	103.10	192.30	283.64	452.46	613.81
135	N/A	79.59	104.14	194.39	285.71	454.29	615.39
140	N/A	79.65	105.06	196.39	287.79	456.23	617.15
145	N/A	79.68	105.84	198.29	289.84	458.24	619.06
150	N/A	79.69	106.53	200.09	291.84	460.29	621.07
155	N/A	79.66	107.11	201.74	293.77	462.35	623.15
160	N/A	79.63	107.59	203.26	295.60	464.39	625.26
165	N/A	79.57	107.99	204.64	297.31	466.36	627.35
170	N/A	79.49	108.32	205.88	298.90	468.25	629.38
175	N/A	79.41	108.58	206.97	300.33	470.02	631.35
180	N/A	79.32	108.76	207.93	301.64	471.67	633.19
185	N/A	79.22	108.90	208.75	302.80	473.19	634.91
190	N/A	79.12	108.99	209.46	303.81	474.55	636.50
195	N/A	79.02	109.02	210.05	304.69	475.76	637.95
200	N/A	78.93	109.02	210.53	305.44	476.84	639.25
205	N/A	78.85	109.00	210.92	306.07	477.78	640.39
210	N/A	78.78	108.93	211.22	306.59	478.57	641.39
215	N/A	78.72	108.84	211.44	307.00	479.25	642.26
220	N/A	78.70	108.73	211.59	307.33	479.80	643.01
225	N/A	78.71	108.60	211.69	307.58	480.28	643.63
230	N/A	78.76	108.46	211.74	307.76	480.64	644.16
235	N/A	78.87	108.31	211.74	307.88	480.94	644.60
240	N/A	79.06	108.14	211.71	307.95	481.16	644.95
245	N/A	79.33	107.98	211.64	307.97	481.33	645.23
250	N/A	79.74	107.81	211.55	307.96	481.44	645.46
255	N/A	80.32	107.65	211.45	307.92	481.52	645.64
260	N/A	81.16	107.49	211.33	307.86	481.57	645.79

TABLE III
TRANSFER TIME, IN DAYS, OF THE CONVERGED PMP-EXTREMAL CANDIDATE TO A CIRCULAR ORBIT OF 1 AU AND FINAL POLAR ANGLE (AT ROCKET ARRIVAL).

Polar Angle (°)	Origin Radius (AU)						
	0.390	0.720	1.520	5.200	9.580	19.220	30.050
0	N/A	N/A	95.77	220.50	337.82	539.56	724.64
5	N/A	N/A	84.93	214.67	331.66	532.46	716.76
10	N/A	N/A	73.02	208.91	325.51	525.31	708.76
15	N/A	N/A	60.90	203.30	319.44	518.16	700.69
20	N/A	N/A	51.21	197.97	313.53	511.08	692.65
25	N/A	N/A	45.92	192.97	307.82	504.14	684.70
30	N/A	32.29	43.85	188.36	302.39	497.40	676.91
35	N/A	28.90	43.65	184.19	297.29	490.93	669.34
40	N/A	28.92	44.59	180.58	292.60	484.76	662.08
45	N/A	29.84	46.15	177.50	288.32	478.95	655.19
50	N/A	31.18	48.07	174.95	284.47	473.58	648.68
55	N/A	32.73	50.23	172.91	281.10	468.66	642.61
60	N/A	34.42	52.55	171.44	278.26	464.19	637.03
65	N/A	36.19	54.97	170.45	275.88	460.24	631.95
70	N/A	37.99	57.46	169.91	273.97	456.77	627.37
75	N/A	39.83	59.98	169.77	272.50	453.80	623.40
80	N/A	41.69	62.51	170.05	271.54	451.38	619.96
85	49.67	43.54	65.05	170.67	270.98	449.42	617.07
90	46.93	45.39	67.57	171.58	270.81	447.98	614.71
95	45.83	47.22	70.08	172.74	270.99	447.01	612.86
100	45.46	49.03	72.54	174.16	271.54	446.45	611.58
105	45.51	50.82	74.97	175.77	272.38	446.33	610.73
110	45.85	52.58	77.36	177.54	273.50	446.58	610.38
115	46.37	54.31	79.67	179.42	274.83	447.17	610.46
120	47.00	55.99	81.93	181.41	276.38	448.10	610.91
125	47.72	57.63	84.12	183.47	278.09	449.30	611.70
130	48.49	59.22	86.23	185.58	279.93	450.74	612.82
135	49.31	60.76	88.27	187.70	281.87	452.36	614.21
140	50.15	62.24	90.22	189.81	283.87	454.17	615.83
145	51.02	63.68	92.07	191.90	285.92	456.10	617.64
150	51.89	65.04	93.84	193.93	287.97	458.09	619.58
155	52.75	66.35	95.49	195.89	289.99	460.14	621.61
160	53.61	67.60	97.05	197.76	291.96	462.19	623.69
165	54.45	68.79	98.49	199.53	293.86	464.22	625.79
170	55.28	69.89	99.84	201.18	295.66	466.18	627.86
175	56.08	70.94	101.08	202.72	297.35	468.07	629.88
180	56.86	71.91	102.23	204.13	298.92	469.85	631.81
185	57.61	72.83	103.26	205.39	300.35	471.50	633.62
190	58.33	73.68	104.18	206.52	301.64	473.03	635.31
195	59.02	74.46	105.02	207.52	302.79	474.40	636.87
200	59.68	75.17	105.76	208.39	303.80	475.63	638.27
205	60.31	75.83	106.41	209.14	304.67	476.72	639.52
210	60.90	76.42	106.97	209.77	305.42	477.67	640.64
215	61.46	76.96	107.45	210.30	306.06	478.48	641.61
220	61.98	77.44	107.87	210.73	306.58	479.17	642.44
225	62.47	77.86	108.21	211.07	306.99	479.75	643.17
230	62.93	78.23	108.48	211.34	307.32	480.22	643.76
235	63.35	78.57	108.69	211.52	307.58	480.60	644.27
240	63.75	78.84	108.84	211.65	307.75	480.90	644.68
245	64.10	79.07	108.95	211.72	307.87	481.14	645.02
250	64.44	79.27	109.01	211.75	307.94	481.31	645.30
255	64.74	79.41	109.03	211.73	307.97	481.43	645.51
260	65.01	79.53	109.01	211.68	307.96	481.51	645.68
265	65.26	79.62	108.97	211.61	307.92	481.57	645.82
270	65.49	79.66	108.88	211.51	307.87	481.60	645.93
275	65.69	79.68	108.78	211.39	N/A	481.61	646.03
280	65.86	79.68	108.66	211.26	N/A	481.62	646.12
285	66.02	79.65	108.52	211.14	N/A	481.62	646.21
290	66.16	79.61	108.36	211.00	N/A	481.63	646.31
295	66.27	79.54	108.21	210.88	N/A	481.66	646.43
300	66.38	79.47	108.04	210.75	N/A	481.71	646.57
305	66.47	79.37	107.87	210.65	N/A	481.78	646.75
310	66.56	79.27	107.70	210.56	N/A	481.89	646.96
315	66.63	79.17	107.54	210.49	N/A	482.04	647.22
320	66.70	79.07	107.39	210.44	N/A	482.24	647.54
325	66.76	78.97	107.25	210.43	N/A	482.48	647.93
330	66.83	78.89	107.13	210.45	N/A	482.79	648.38
335	66.90	78.80	107.03	210.51	N/A	483.16	648.91
340	66.99	78.74	106.97	210.60	N/A	483.61	649.55
345	67.08	78.70	106.94	210.75	N/A	484.14	650.27
350	67.19	78.70	106.94	210.96	N/A	484.76	651.11
355	67.32	78.73	107.00	211.22	N/A	485.47	652.06
360	67.48	78.82	107.10	211.54	N/A	486.28	653.13

TABLE IV
TRANSFER TIME, IN DAYS, OF THE CONVERGED PMP-EXTREMAL CANDIDATE TO A CIRCULAR ORBIT OF 1 AU AND INITIAL POLAR ANGLE (AT ROCKET DEPARTURE).

Polar Angle ($^{\circ}$)	Origin Radius (AU)						
	0.390	0.720	1.520	5.200	9.580	19.220	30.050
-100	46.94	N/A	137.22	281.20	401.02	604.64	789.45
-95	46.28	N/A	133.71	279.71	400.45	605.10	790.72
-90	45.79	N/A	130.04	277.91	399.53	605.19	791.59
-85	45.47	N/A	126.23	275.81	398.24	604.86	792.02
-80	45.48	N/A	122.29	273.35	396.55	604.09	791.99
-75	45.94	N/A	118.16	270.58	394.50	602.83	791.40
-70	47.06	N/A	113.88	267.49	392.03	601.11	790.28
-65	49.09	N/A	109.49	264.09	389.19	598.93	788.57
-60	51.87	N/A	104.93	260.40	385.96	596.20	786.30
-55	54.64	N/A	100.21	256.39	382.37	593.01	783.38
-50	56.91	N/A	95.40	252.12	378.37	589.32	779.89
-45	58.67	N/A	90.40	247.61	374.03	585.12	775.84
-40	60.08	42.22	85.27	242.85	369.36	580.46	771.21
-35	61.20	39.24	80.06	237.91	364.41	575.38	766.07
-30	62.11	36.39	74.72	232.81	359.17	569.87	760.34
-25	62.88	33.74	69.36	227.59	353.71	564.01	754.16
-20	63.51	31.37	64.05	222.28	348.05	557.81	747.56
-15	64.05	29.82	58.75	216.95	342.25	551.33	740.58
-10	64.50	28.85	53.84	211.66	336.37	544.62	733.26
-5	64.88	29.29	49.74	206.47	330.45	537.73	725.67
0	65.21	31.36	46.31	201.42	324.55	530.74	717.89
5	65.49	34.83	44.22	196.60	318.73	523.70	709.96
10	65.72	39.16	43.58	192.04	313.06	516.67	701.98
15	65.92	43.77	44.15	187.81	307.57	509.73	694.00
20	66.09	48.26	45.83	183.96	302.33	502.94	686.10
25	66.23	52.45	48.22	180.57	297.40	496.34	678.35
30	66.35	56.21	51.07	177.63	292.84	490.00	670.82
35	66.46	59.57	54.25	175.16	288.65	483.95	663.55
40	66.54	62.47	57.61	173.14	284.87	478.29	656.62
45	66.63	65.02	61.04	171.60	281.51	473.04	650.08
50	66.70	67.22	64.50	170.57	278.64	468.22	643.97
55	66.77	69.13	67.94	169.97	276.22	463.84	638.28
60	66.84	70.77	71.32	169.78	274.27	459.96	633.13
65	66.92	72.19	74.61	169.97	272.76	456.55	628.48
70	67.01	73.42	77.78	170.54	271.68	453.63	624.35
75	67.12	74.48	80.81	171.43	271.05	451.26	620.77
80	67.24	75.39	83.70	172.59	270.82	449.33	617.75
85	67.41	76.18	86.42	173.99	270.94	447.93	615.28
90	67.62	76.85	88.98	175.64	271.40	446.97	613.33
95	N/A	77.43	91.36	177.44	272.19	446.44	611.88
100	N/A	77.92	93.56	179.39	273.28	446.33	610.94
105	N/A	78.33	95.57	181.45	274.59	446.60	610.43
110	N/A	78.67	97.42	183.58	276.10	447.20	610.38
115	N/A	78.95	99.07	185.77	277.82	448.15	610.75
120	N/A	79.18	100.57	187.97	279.67	449.37	611.47
125	N/A	79.36	101.92	190.15	281.62	450.82	612.50
130	N/A	79.49	103.11	192.30	283.64	452.46	613.81
135	N/A	79.59	104.14	194.39	285.71	454.29	615.39
140	N/A	79.65	105.06	196.39	287.79	456.23	617.15
145	N/A	79.68	105.85	198.29	289.84	458.24	619.06
150	N/A	79.69	106.54	200.09	291.84	460.29	621.08
155	N/A	79.66	107.11	201.74	293.77	462.35	623.15
160	N/A	79.63	107.60	203.26	295.60	464.39	625.26
165	N/A	79.57	108.00	204.64	297.31	466.36	627.35
170	N/A	79.49	108.32	205.88	298.90	468.25	629.38
175	N/A	79.41	108.58	206.99	300.34	470.02	631.35
180	N/A	79.32	108.76	207.93	301.65	471.67	633.20
185	N/A	79.22	108.90	208.75	302.80	473.19	634.91
190	N/A	79.12	108.98	209.46	303.82	474.54	636.50
195	N/A	79.02	109.02	210.05	304.69	475.76	637.95
200	N/A	78.93	109.03	210.52	305.45	476.84	639.25
205	N/A	78.85	108.99	210.92	306.08	477.77	640.40
210	N/A	78.77	108.93	211.22	306.59	478.57	641.40
215	N/A	78.72	108.84	211.44	307.01	479.25	642.26
220	N/A	78.70	108.73	211.59	307.34	479.81	643.01
225	N/A	78.71	108.60	211.70	307.59	480.28	643.64
230	N/A	78.76	108.46	211.74	307.76	480.64	644.17
235	N/A	78.87	108.30	211.74	307.88	480.93	644.60
240	N/A	79.06	108.15	211.71	307.94	481.17	644.95
245	N/A	N/A	107.97	211.64	307.97	481.33	645.24
250	N/A	N/A	107.81	211.56	307.96	481.44	645.47
255	N/A	N/A	107.65	211.45	307.92	481.52	645.64
260	N/A	N/A	107.49	211.33	307.87	481.57	645.79